

EXHIBIT F

**IN THE UNITED STATES DISTRICT COURT FOR THE
DISTRICT OF COLORADO**

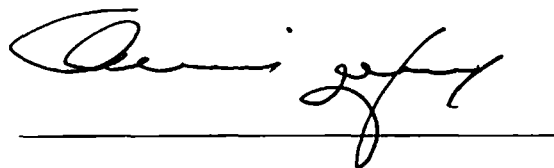
United States of America

v.

*CEMEX, Inc.
Lyons Cement Plant
Lyons, Colorado*

Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

**EXPERT REBUTTAL REPORT
OF
PHYLLIS FOX, Ph.D., PE**

A handwritten signature in black ink, appearing to read 'Phyllis Fox', is written over a horizontal line.

June 24, 2011

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I. BACKGROUND

A. Information Required by Federal Rules of Civil Procedure

1. This Rebuttal Report contains my opinions, conclusions, and the reasons therefore.
2. Appendix A to the Expert Report of Phyllis Fox, Ph.D., PE, March 1, 2011, contains a statement of my qualifications and publications.
3. Appendix A affixed hereto lists the references, data, and other information I considered in forming the opinions I express in this Rebuttal Report.
4. Appendix C to the Expert Report of Phyllis Fox, Ph.D., PE, March 1, 2011, contains a list of my previous testimony as an expert witness at trial or by deposition in the last four years.
5. A summary of my experience is found at pages 3 – 4 of the Expert Report of Phyllis Fox, Ph.D., PE, March 1, 2011.
6. I am being compensated at a rate of \$200/hr for my work in this matter.

The opinions expressed in this Rebuttal Report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

B. Areas of Requested Opinion

I have been asked by the United States Department of Justice (DOJ) to review the reports of Messrs. John E. Hofmann,¹ Gerald L. Young,² and Dr. Kevin Davis³ tendered on behalf of CEMEX, Inc. (CEMEX) in this matter on May 10, 2011 and to identify and rebut certain assertions made by those gentlemen that I disagree with.

¹ Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011 (hereinafter “Hofmann Expert Report”).

² Expert Report of Gerald L. Young, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011 (hereinafter “Young Expert Report”).

³ Expert Report of Kevin Davis, Ph.D., United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011 (hereinafter “Davis Expert Report”).

II. EMISSION FACTORS

An emission factor is the amount of a pollutant that is generated per unit of raw material or product. NOx emissions from cement kilns are calculated using an emission factor expressed in pounds of NOx per ton of clinker (or raw feed) (“lb/ton”).⁴ Annual NOx emissions in tons per year are calculated by multiplying this emission factor by the tons per year of clinker. Thus, an increase in clinker, by simple arithmetic,⁵ increases NOx emissions. In general, making more clinker requires burning more fuel, which increases NOx emissions.⁶ However, CEMEX’s experts argue the opposite, that the NOx emissions in lb/hr and lb/ton decline as feed and fuel input increase.

I assumed that the pre- and post-change emission factors in pounds of NOx per ton of clinker were equal, *i.e.*, that the projects did not alter the ratio of NOx to clinker. Mr. Hofmann argues that “it is inappropriate to utilize a constant emission factor based solely on clinker production for evaluating PSD applicability for the projects at issue in this case.”⁷ Elsewhere, he argues that the NOx emission factor “is not at all constant over the range of clinker production” and “utilizing the NOx emission factor constant over a long period of time ignores the NOx reductions and other improvements of a project.”⁸

Dr. Davis opines that “a simple application of emission factors would be of little value in evaluating potential NOx impacts related to the tertiary air oxygen enhancement system and calciner geometry improvements...,” due to the complexity and sensitivity of NOx emissions, the magnitude and nature of the fluctuations in historical NOx data, and the unique combustion effects of the calciner improvements.⁹

Neither expert offers an alternative approach for evaluating PSD applicability in the pre-construction phase of the subject projects. Moreover, the theories they do present are inconsistent with the analysis CEMEX conducted contemporaneously with these projects and CEMEX’s own permitting practices.

⁴ U.S. Environmental Protection Agency (U.S. EPA), AP-42, Fifth Edition, January 1995, Compilation of Air Pollutant Emission Factors, Vol. 1, Stationary Point and Area Sources, Section 11.6, Portland Cement Manufacturing, Table 11.6-8, Emission Factors for Portland Cement Manufacturing (hereinafter, “U.S. EPA, AP-42”).

⁵ Deposition of Andrew Jackura, September 16, 2010, TR at 116:7 – 10.

⁶ Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011 (hereinafter “Greer Expert Report”), p. 27; Deposition of Ram Seetharam, August 11, 2010, TR at 58:5 – 59:11, 107:15 – 20, 110:7 – 9.

⁷ Hofmann Expert Report, p. 1-4, Opinion 6.

⁸ Hofmann Expert Report, p. 2-10.

⁹ Davis Expert Report, p. 2, Opinion 5, and p. 17 (“Based on fundamental consideration, the two modifications discussed herein could reasonably be expected to reduce NOx emissions while increasing fuel efficiency and clinker production.”).

A. Constant Emission Factor Is Conservative

CEMEX's experts allege that I should not have used a constant emission factor to estimate the increase in NOx emissions for the projects at issue in this case. Rather, they argue the types of projects that were implemented improved efficiency, which they contend would have reduced the emission factor.¹⁰ However, CEMEX's experts present no evidence to support this proposition.

CEMEX itself analyzed the trend in emission factors between 1995 and 2006 in a May 2007 Environmental Strategic Plan.¹¹ CEMEX's analysis showed a sharp increase in the NOx emission factor from 1997 to 2001. I prepared a similar analysis using the data compiled by CEMEX for this case, corrected for missing data in 1998 and 1999¹² and for CEMEX's improper moisture correction in 2000 and 2001.¹³ My analysis is discussed in Section III and summarized in Figure 1 in pounds per ton clinker (lb/ton).

¹⁰ Hofmann Expert Report, p. 1-4, Opinion 6, and p. 2-9; Davis Expert Report, p. 17.

¹¹ CEMEX, Environmental Strategic Plan, Lyons Plant, May 2007, CMXLYO00151454 at 151464 (slide 11) – 151501; Annual Emissions Inventory, Environmental Strategy, Lyons 2007, CMXLYO00151502 – 151505. *See also*, Annual Emissions Inventory, Environmental Strategy, Lyons 2007, CMXLYO00113389.

¹² *See* Section III of this report.

¹³ In its CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845, CEMEX inappropriately reduced its estimates for NOx emissions for the years 2000 and 2001 by approximately 19%. *See*, Rebuttal Expert Report of Peter R. Westlin, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 15, 2011 (hereinafter “Westlin Rebuttal Report”), p.3 (“The DAS used with the CEMEX CEMS for most of the 2000 through 2001 period appropriately corrected for stack moisture content.”). *See also*, Westlin Rebuttal Report, p. 16 (“That means that for at least the first 18 months (and probably the first 22 months) of the 2000-2001 period, the NOx mass emissions data produced by the CEMEX CEMS was appropriately correcting for stack gas moisture content and any further adjustments for moisture would be unnecessary and improper.”).

Figure 1
Trend in NOx Emission Factor (lb/ton)

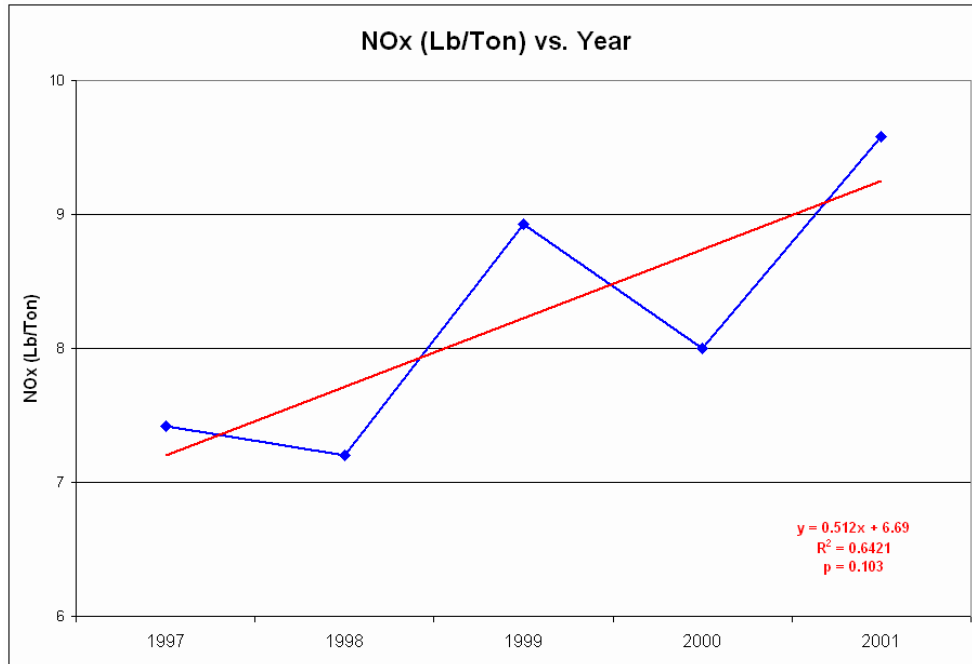


Figure 1 shows that the NOx emission factor in pounds per ton of clinker, the metric I used, increased from 7.6 lb/ton in 1997 to 9.2 lb/ton in 2001, the years when the expansion projects were brought on line. A constant emission factor is consistent with the long-term history at Lyons.¹⁴ As Figure 1 illustrates, the NOx emission factor went up over the subject period. Thus, my use of a constant 7.6 lb/ton emission factor potentially underestimates NOx emissions and is conservative in favor of CEMEX.

B. Short-Term Variability Is Not Relevant for PSD Applicability

Mr. Hofmann argues that it is difficult to accurately predict NOx emissions over short durations and to quantify the short-term NOx emissions associated with a particular change. “The identification of a single change and its individual impact to overall NOx emissions is difficult at best.”¹⁵ I agree that NOx emissions from cement kilns are highly variable over short durations.¹⁶ In fact, this is one of the reasons that the 1997 and 1998 oxygen tests are unreliable. Mr. Hofmann contradicts himself, first by arguing it is difficult to identify the impact of a single

¹⁴ Expert Report of Phyllis Fox, Ph.D., PE, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011 (hereinafter “Fox Expert Report”), p. 18 – 19.

¹⁵ Hofmann Expert Report, p. 2-6.

¹⁶ Fox Expert Report, p. 18.

change on NO_x due to short-term variability (from 1 hour to less than 30 days)¹⁷ and then by arguing oxygen injection caused the NO_x emission factor to decline. The oxygen tests that Mr. Hofmann relied on to reach this latter conclusion are based on short-term (1-hour) measurements of NO_x and oxygen.¹⁸

Short-term variability is not a major factor at the scale of a PSD applicability determination. The potential to emit used in these calculations is based on a two-year annual average, which smoothes out short-term variability. Variability on the scale of hours, days, and even months is irrelevant for PSD analysis, which is based on annual changes. In 2002, Mr. Young, one of CEMEX's experts, explained that based on NO_x CEMS data, at least 14 to 30 days of data were required to accurately estimate NO_x emissions:

Data are available from a tightly controlled, well-documented research project concerning NO_x emissions from a long dry process kiln in southern California (Young et. al, 1986). These data demonstrated the highly variable nature of NO_x emissions. When these NO_x data were subjected to statistical analyses, it was determined that for the estimated mean emissions rate to be statistically accurate within 10 percent of the true emissions rate, 30.7 days of emissions data were required. Similar tests conducted on a preheater kiln determined that about 14 days of emissions data were required for similar statistical accuracy (Young and von Seebach, 1999).¹⁹

This supports my contention that the 2-year annual averages used in my netting analyses yield accurate estimates of NO_x emissions. This further supports the proposition that the 1997 and 1998 oxygen tests, which are based on 1-hour measurements of NO_x and oxygen, do not accurately predict NO_x emissions. Thus, while NO_x emissions are variable, if enough data is collected, *e.g.*, the two-year averages I used, an accurate estimate of the emissions rate can be obtained. The 1-hour averages used in the 1997 and 1998 oxygen tests are not long enough to accurately estimate NO_x emissions or other variables.

The emissions data that I used to estimate the net increase in NO_x emissions is based on two full years of continuous emission monitoring data from the CEMEX Lyons facility. Contrary to Mr. Hofmann's implication, my analyses did not involve using canned emission factors, predicting NO_x emissions over short durations, or quantifying short-term emissions associated with a particular change.²⁰ Rather, in my March 1, 2011 Expert Report, I used long-

¹⁷ Hofmann Expert Report, p. 2-6 (High variability is the reason many kilns have NO_x emissions limits averaged over a month or longer time period than an hour.).

¹⁸ Hofmann Expert Report, Sec. 5.3, beginning at p. 5-10.

¹⁹ Gerald L. Young, NO_x Formation in Rotary Kilns Producing Cement Clinker Applicable NO_x Control Techniques and Cost Effectiveness of these Control Techniques, IEEE Environmental, Safety and Maintenance Working Group, Jacksonville Conference May 2002, p. 243.

²⁰ Hofmann Expert Report, p. 2-6:

term, annual average emissions that I calculated from 2 years of site-specific data, using the standard potential to emit methodology.

C. Site-Specific Emission factors Should Be Used

Mr. Hofmann supports his variability argument by pointing to emission factors published in various reports, noting that they cover a broad range and concluding that site-specific data should be used to determine emission factors.²¹ However, none of the material he points to is specific to the Lyons kiln.²² He then concludes, “In my experience, NOx emissions from a cement kiln similar to Lyons would rely on site-specific emission data (stack testing/CEMS).”²³ I agree with Mr. Hofmann on this point and note that my netting analyses – unlike Mr. Hofmann’s discussion of NOx variability – were based on site-specific emission factors that I calculated from 2 years of Lyons CEMS data.²⁴

D. A Constant Emission Factor Is Standard Practice to Estimate Potential to Emit

Mr. Hofmann argues that “one must account for inherent variability or project-specific improvements in NOx emissions, to evaluate any changes in NOx.”²⁵ He claims that relying on a “fixed lb/ton emission factor without considering project-specific NOx related reductions or improvements” is problematic, as this approach “does not consider any efficiency and improvements type of projects” and “does not consider inherent variability and factors controlling NOx emissions, which are not solely dependent on clinker production.”²⁶ Dr. Davis echoes this argument, stating: “a simple consideration of clinker production and application of [a] constant emission factor is not a reasonable approach to make conclusions regarding NOx emissions due to equipment modifications to the Lyons cement kiln.”²⁷ However, Mr. Hofmann and Dr. Davis cite no record evidence, authority, or guidance and provide no engineering

The inability of the kiln operator to keep kiln conditions, and thus NOx emissions, constant results in a difficulty to accurately [sic] predict NOx emissions over short durations and to quantify the short-term NOx emissions associated with a particular change.... The identification of a single change and its individual impact to overall NOx emissions is difficult at best.

²¹ Hofmann Expert Report, p. 2-7 to 2-8, Tables 2-1 and 2-2.

²² Hofmann Expert Report, p. 2-7 to 2-8, Tables 2-1 and 2-2.

²³ Hofmann Expert Report, p. 2-8.

²⁴ Fox Expert Report, p. 12 – 14,²⁵ Hofmann Expert Report, p. 2-8.

²⁵ Hofmann Expert Report, p. 2-8.

²⁶ Hofmann Expert Report, p. 2-9.

²⁷ Davis Expert Report, p. 18.

analyses to support their contention that the application of a constant emission factor is unreasonable under the circumstances here. Further, neither of them provides a fully developed alternative that could have been employed during the relevant pre-construction time period to predict potential post-construction emissions changes.

Two pieces of information are required to estimate the increase in emissions from a change: an estimate of the emissions before the change (baseline); and an estimate of the emissions after the change. I estimated the emissions after the change as the potential to emit. I calculated an emission factor in pounds NO_x per ton clinker for the baseline years and multiplied it by the increase in clinker production anticipated by CEMEX. This essentially assumes that the emission factor remains constant, or “fixed,” between the baseline and the future. This is a standard practice that CEMEX itself has relied upon. As CEMEX’s Corporate Director of Air Permitting testified in the Victorville case:

Q: Now, does the day-in/day-out use of a constant NO_x emission factor—does that assume that there is a relationship, then, between production and emissions?
A: No. Q: So, the emission factor appears to stay constant in your table. A: Yes.
Q: So, how is it that there is not—I mean, all you’re doing is saying emission factor times production equals emissions. A: That is correct. Q: So, in that calculation, as production goes up, emissions go up. Is that correct? A: That is correct.²⁸

The PSD program is a preconstruction permitting program. The “potential to emit” must be estimated before the project is built to determine if PSD applies. The potential to emit is the maximum design capacity to emit a pollutant, except as constrained by federally enforceable conditions.²⁹ A proponent of a project that has the potential to increase emissions above the significance threshold must predict post-project emissions, install pollution controls, or acceptable a federally enforceable limit. As CEMEX did not install any pollution controls or accept a federally enforceable NO_x limit, its only remaining alternative was to calculate post-project potential emissions to determine whether the changes it was planning to make would trigger PSD.

The standard method to predict post-project potential emissions is to make an estimate using an emission factor.³⁰ EPA’s “Compilation of Air Pollutant Emission Factors,” released in

²⁸ United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Deposition of Amarjit Singh Gill, March 18, 2008, TR at 362:21 – 363:13.

²⁹ 40 CFR 52.21(b)(4) (1995) (presently recodified as 40 CFR 52.21(b)(23)(i)); Colorado Air Quality Control Commission (CAQCC), Common Provisions, I.G (1986); CAQCC Part A, I.B.44 (1997). *See also*, New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting, Draft, October 1990 (hereinafter “NSR Manual”), p. A.1 and Sec. II.B, p. A.4 – A.22.

³⁰ U.S. EPA, AP-42, Sec. 11.6, Portland Cement Manufacturing, Table 11.6-8, Emission Factors for Portland Cement Manufacturing; U.S. EPA, Alternative Control Techniques Document – NO_x Emissions from Cement Manufacturing, Report EPA-453/R-94-004, March 1994. Available at: <http://www.epa.gov/ttn/catc/dir1/cement.pdf>; NSR Manual, Section II.B.6 and Appendix C.

January 1995, before these changes were implemented, explains: “Emission factors may be appropriate to use in a number of situations.... Emission factor use may also be appropriate in some permitting applications, such as in applicability determinations and in establishing operating permit fees.”³¹ The Florida Department of Environmental Protection, for example, required CEMEX to use the (baseline) NOx emission factor expressed in pounds of NOx per ton of dry feed to estimate potential to emit. The final permit was issued based on this emission factor in lb/ton dry feed.³²

The firm retained by CEMEX to permit the Lyons quarry and separator was McVehil-Monnett Associates (MMA).³³ The quarry and separator are the only part of the expansion project that was permitted. See Section VIII. In pursuing permitting of the quarry and separator, MMA determined that the kiln, cooler, and dryer should be permitted together with the quarry and separator and so advised CEMEX.³⁴ MMA then calculated the NOx emission factor for the kiln from the anticipated increase in raw feed and a baseline emission factor in pounds of NOx per ton of raw feed, based on 1997 CEMS data.³⁵ MMA’s estimates of the net increase in NOx emissions ranged from 324 ton/yr³⁶ to 452 ton/yr³⁷ and agree with my estimate (465 – 471 ton/yr).³⁸ I found no evidence in the documents produced to date or in deposition testimony that

³¹ U.S. EPA, AP-42, Introduction, p. 2.

³² Memorandum from Al Linero to Howard Rhodes, Re: Southdown, Inc., 0530010-003-AC and PSD-FL-233, Modification of Existing Construction Permit, Final Permit, June 17, 1997, Attaching Notice of Final Permit, Final Determination, and Air Construction Permit, pdf 32 (BACT limits) and Table 1-2.

³³ Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, Re: Lyons Kiln Production and Oxygen Use, January 4, 1998, CMXLYO00039059 – 39061.

³⁴ Deposition of Keith Baugues, July 30, 2010, TR at 30:22 – 31:8. “Q. Okay. Did CEMEX ask you to look at kiln, the clinker cooler, the dryer? A. I don’t believe they asked me to do that. That is something I did independently. Q. And did you advise CEMEX that these units should be included in a permit application? A. Yes, I did. Q. Why did you feel they should be included? A. I felt at that point that the increases in capacity would increase the emissions from those units.”

³⁵ Deposition of Keith Baugues, July 30, 2010, TR at 91:17 – 95:3. *See also*, Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on “Kiln and Dry Emissions,” US00000065 at 80 – 82, US00000062 (conclusions); US00000090 (Table 1 handwritten); US00034559 at 34562 (fax cover sheet); US00034618 – 34620 (memo); and US00034473 at 34474 (Table 1 typewritten). Letter from Keith Baugues to Bradley Cooley, Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294.

³⁶ Emissions Summary (TPY), 1996 – 97 Baseline, US00028915 at 28921 (786-98/Keith/Summary.xls) and US00000047 at 52 (**323.63 ton/yr**); Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on “Kiln and Dry Emissions,” US00000065 at 80 – 82, and US00000090 (Table 1 handwritten, **365 ton/yr**); Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 at 38874 (**365 ton/yr**).

³⁷ Letter from Keith Baugues to Bradley Cooley, Southdown, Inc., Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294; Emissions Summary (TPY), 1995 – 96 Baseline, US00028915 at 28920 (786-98/Keith/Summary.xls) and US00000047 at 51 (**451.93 ton/yr**); Deposition of Keith Baugues, July 30, 2010, TR at 158:21 – 23 (“I believe these were final calculations before we prepared the permit application.”).

³⁸ Fox Expert Report, p. 20 – 21.

CEMEX rejected the MMA constant emission factor or netting calculations that relied on it.³⁹ See further discussion of the MMA netting analysis in Section VIII.C.

E. CEMEX's Suggested Reduction in NOx Emission Factors is Inappropriate

Mr. Hofmann and Dr. Davis both suggest that the NOx emission factors I used to calculate emissions after the projects were built should be reduced to account for NOx-related reductions such as improvements in efficiency.⁴⁰ They do not provide any information on how one might go about this nor do they cite to any place in the record that even suggests CEMEX contemplated or attempted to undertake such calculations in the pre-construction period.

First, the suggestion that the NOx emission factor should be lowered is inconsistent with the applicable emissions test. The PSD regulations state that “[f]or any emissions unit which has not begun normal operations on the particular date, actual emissions shall equal the potential to emit of the unit on that date.”⁴¹ The potential to emit is the capability of a source at maximum design capacity to emit a pollutant, except as constrained by federally enforceable conditions.⁴² Potential to emit is the appropriate method to estimate post-construction emissions from the subject changes at the time they should have been permitted.⁴³

³⁹ Deposition of Keith Baugues, July 30, 2010, TR at 250:9 – 16 (“Q: Okay, With respect to the netting analysis, particularly as to the kiln and dryer, did anyone at the plant ever tell you that that analysis was erroneous? A: No. Did they accept that analysis as being a valid analysis? A: Yes.”).

John Lohr testified that Jeet Gill criticized the Baugues analysis as being too “simplistic.” Mr. Lohr incorrectly testified the analysis was based on AP-42 emission factors when, for the kiln, it was based on actual Lyons CEMS emissions data, the most accurate way to estimate emissions. Deposition of John Lohr, October 11, 2010, TR at 184:16 – 185:16, 186: 13 – 15, and 187:3 – 15.

Deposition of William Monnett, November 30, 2010, TR at 29:5 – 10 (“Q: Let me ask you this: Did CEMEX ever come to you and say that they disagreed with the emission factors Mr. Baugues used in this analysis? A: I don't believe they did. Not that I recall.”), 33:15 – 25, 51:24 – 52:9 ([Jeet Gill never informed McVehil-Monnett that their netting analysis was wrong]: “...I recall him saying at a meeting, I don't care what your opinion is, I have mine.”).

Deposition of Bradley Cooley, September 20, 2010, TR at 37:16 – 19, 97:18 – 20 (“Q: Did you ever hear anyone at CEMEX criticize any netting analysis done by McVehil-Monnett? A: No, not specifically.”), 98:8 – 10.

⁴⁰ Hofmann Expert Report, p. 2-8 to 2-10; Davis Expert Report, p. 17 – 18.

⁴¹ 40 CFR 52.21(b)(21)(iv); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.1.c (1997).

⁴² 40 CFR 52.21(b)(4) (1995) (presently recodified as 40 CFR 52.21(b)(23)(i)); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.44 (1997). *See also*, NSR Manual, p. A.1 and Sec. II.B, p. A.4 – A.22.

⁴³ *See*, NSR Manual, p. A.39 and A.41; Deposition of Keith Baugues, July 30, 2010, TR at 249:11 – 15.

CEMEX's experts characterize these changes as efficiency improvements.⁴⁴ (However, at the time they were implemented, they were characterized and sold to management as capacity increases.) In other words, CEMEX's experts assert after the fact that these modifications would allow the production of the same amount of clinker while burning less fuel. Because burning fuel generates NOx, under CEMEX's theory, the less fuel burned, the less NOx generated while producing the same amount of clinker. CEMEX asserts this caused the emission factor, in pounds per ton clinker, to go down.

An improvement in efficiency can be realized in three ways: (1) the same amount of clinker could be made by burning less fuel, which would reduce NOx emissions in lb/hr; (2) more clinker could be made using the same amount of fuel; (3) or, more clinker could be made by burning more fuel, up to the capacity of the system, which would increase NOx emissions in lb/hr.⁴⁵

An improvement in efficiency always has the potential of increasing production and hence increasing NOx emissions, unless there is a physical or federally enforceable limit on throughput or emissions.⁴⁶ Without a throughput limitation, CEMEX could decide at any point in the future to operate the more efficient equipment at a higher throughput, up to its physical capacity, producing more clinker.

As discussed below in Section III, CEMEX implemented these projects not to make the same amount of cement by burning less fuel but rather to capture the benefit of any efficiency

⁴⁴ Hofmann, p. 1-4, Opinion 6; p. 1-7, Opinion 22; p. 5-3. Young, p. 5 – 6, Opinion 1; p. 7, Opinion 7; Davis, p. 2 – 3, Opinion 1 – 3, 6.

⁴⁵ CEMEX chose the latter alternative.

⁴⁶ See *Puerto Rican Cement Co. v. U.S. EPA*, 889 F.2d 292 (1st Cir. 1989) (upholding EPA's authority to regulate potential to emit at the maximum capacity for an efficiency improvement project.) See, e.g.:

EPA has simply taken account of, and given controlling weight to, a different consideration: the fact that a firm's decision to introduce new, more efficient machinery may lead the firm to decide to *increase the level of production*, with the result that, despite the new machinery, overall emissions will increase. Indeed, EPA points out that a firm introducing such machinery can escape PSD review simply by promising that it will ensure its actual emissions do not in fact increase (that is, by promising that it will not run the machinery at such a rate as to create an actual increase in emissions levels).

Id. at 297.

The Company is not interested in peak load capacity; it operated its old kilns at low levels in the past; its new, more efficient kiln might give it the economic ability to increase production; consequently, EPA could plausibly fear an increase in actual emissions were it to provide the NAD [non-applicability determination]. Thus, this seems the very type of case for which the regulations quoted above were written. We can find nothing arbitrary or irrational about EPA applying those regulations to the Company's proposal.

Id. at 298.

improvement by burning more fuel to make more cement to sell into an over-sold market.⁴⁷ These projects are described throughout the record as capacity increase projects. They were variously called the “Lyons **Capacity Increase**”;⁴⁸ “Lyons Plant **Expansion**”;⁴⁹ “Phase I **Expansion**”;⁵⁰ “Plant **Capacity Increase**”;⁵¹ “Plant **Expansion** Project”;⁵² and “Lyons **Expansion** Project,” among others, in CEMEX internal documents. CEMEX was successful in implementing this approach, as clinker production increased from 419,000 tons in 1994⁵³ to 539,992 tons in 2000.⁵⁴ This data is plotted in Figure 2.

⁴⁷ See, e.g., Lyons Capacity Increase, October 17, 1997, CMXLYO00182398 (Calculation showing how an efficiency improvement to 4.4 MMBtu/ton is equivalent to a 30% production increase); Text and Figures for FCT AFE, CMXLYO00045595.

The Lyons kiln is currently limited by permit to a fixed amount of fuel and feed. Any improvements in efficiency could be expected to be translated into annual capacity up to a maximum of 600,000 cement tons per year (570,000 clinker tons). Current efforts with oxygen injection and mix optimization are expected to push system capacity to about 535,000 clinker or 565,000 cement leaving room for up to 35,000 tons of cement capacity.

Text and Figures for FCT AFE, CMXLYO00045595 (emphasis added).

⁴⁸ Memorandum from Karen Krenak to Norris Widener, November 18, 1998, Re: Approved AFE, US00000454 at 454, 455.

⁴⁹ *Id.* at 456, 457, 462.

⁵⁰ Southdown, Inc., Authorization for Expenditure, November 18, 1998, US000000027 at 37 – 38; Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, CEMEX_NEIC_00003247 at 3251.

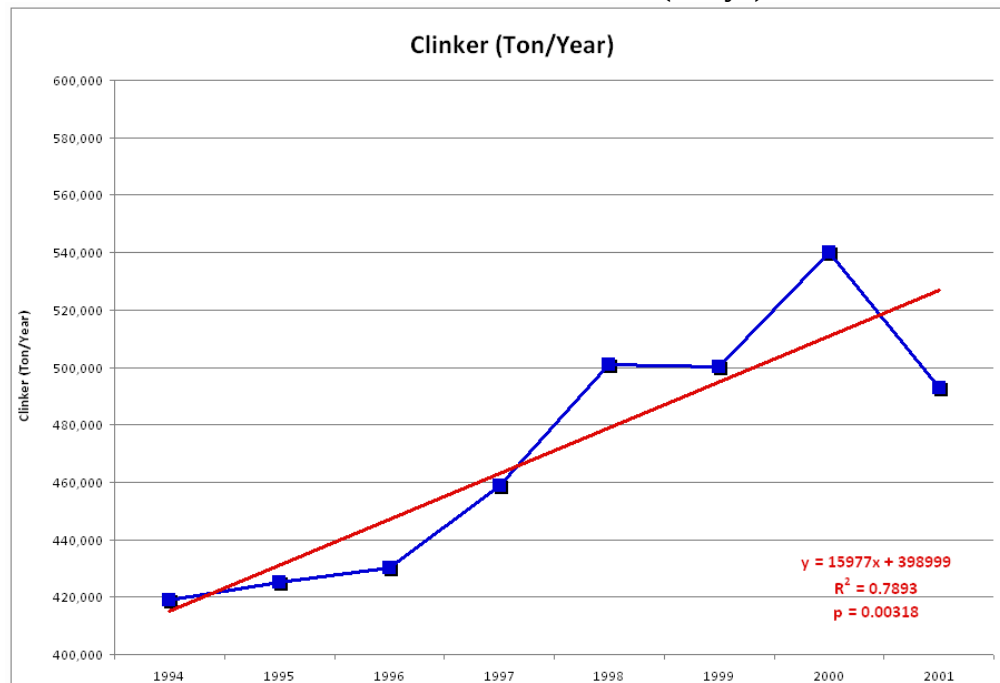
⁵¹ Letter from David Repasz, Southdown, Inc., to Kenneth M. Hitcho, Humboldt Wedag, Inc., June 12, 1998, Re: Letter of Intent for Supply to Finish Grinding Separator System for Lyons, Colorado Plant Authorized Spending, CEMEX_NEIC_00003381 at 3382.

⁵² Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, CEMEX_NEIC_00003247 at 3248.

⁵³ Southdown, Inc., Lyons Cement Plant, Clinker Production, US00000583; Southdown, Inc., Lyons Cement Plant, Air Pollution Emission Notice, December 6, 1995, US00006692 at 6694.

⁵⁴ CEMEX Lyons Facility, Estimated Clinker Production – Historical Summary, CMXLYO00628849 at 628850.

Figure 2⁵⁵
Trend in Clinker Production (ton/yr)



Thus, CEMEX utilized any efficiency improvements to increase throughput, not to reduce NOx emissions by maintaining throughput at the pre-modification level.⁵⁶ As CEMEX did not accept a federally enforceable limit on throughput and did in fact ramp up production (Figure 2), contrary to Mr. Hofmann's suggestions, there is no basis to reduce the NOx emission factor. Such a reduction is inconsistent with the potential to emit test in this case and could only be contemplated under the future actual test, which does not apply to these changes.

Second, while I agree that some types of changes could lead to a reduction in the NOx emission factor,⁵⁷ the types of changes in this case (even if they were valid for a potential to emit test) were too small to detect, as discussed in the next Section. Thus, the improvements in the NOx emission factor likewise would be too small to detect.

Finally, CEMEX's arguments about variability contradict the suggestion that I should have lowered the NOx emission factor. The CEMEX variability argument suggests it is not feasible to determine individual project changes. Dr. Davis, for example, claims: "[I]t would be illogical to make any assumptions regarding the impact of individual changes in operation, fuel,

⁵⁵ Plotted data from CEMEX Lyons Facility, Estimated Clinker Production – Historical Summary, CMXLYO00628849, supplemented by Fox Expert Report, Exhibit 1, p. 50.

⁵⁶ See discussion of debottlenecking in Section VI.

⁵⁷ *E.g.*, the installation of pollution control equipment.

raw mix, or equipment based on simple emission factors.”⁵⁸ He has essentially said that due to variability of Lyons NOx emission data, it is impossible to determine the impact of any given “efficiency project” on NOx emission factors.

III. EFFICIENCY

Efficiency is the amount of energy it takes to make a given quantity of product. For cement plants, it is generally expressed in units of millions of BTUs per ton of clinker (“MMBtu/ton”). Theoretically, if the amount of fuel required to generate one ton of clinker is lowered, the efficiency is improved, *i.e.*, the MMBtu/ton is reduced, and NOx emission would go down as less fuel would be needed to make the same amount of clinker.

The CEMEX experts argue that the 1997 – 2001 changes improved efficiency,⁵⁹ which they contend resulted in a decrease in overall NOx emissions and the NOx emission factor. Mr. Hofmann argues that the subject changes “increased the system’s efficiency and production capacity with no changes in NOx emissions.”⁶⁰ He specifically argues the calciner combustion chamber modifications⁶¹ and booster fan⁶² changes improved efficiency, reducing NOx emissions. Mr. Young concludes that “the efficiency improvements to the calciner and the use of OE together resulted in an increase in clinker production but a decrease in annual NOx emissions.”⁶³ Mr. Young also argues the combustion chamber modifications⁶⁴ and the annulus

⁵⁸ Davis Expert Report, p. 17.

⁵⁹ They do not define efficiency, which they variously called energy efficiency, thermal efficiency, fuel efficiency, and efficiency. These are all presumed to be MMBtu/ton.

⁶⁰ Hofmann Expert Report, p. 1-4, Opinion 6. *See also, id.* at p. 2-10 (criticizing U.S. experts for using an emission factor “without adjusting...for the efficiency improvement projects...”), p. 5-25 (“Improved stability should improve efficiency/reduce NOx emissions.”).

⁶¹ Hofmann Expert Report, p. 1-4, Opinion 10:

This project was implemented as an improvement to the calciner combustion and allowed CEMEX to more effectively utilize the thermal output of the fuel injected into the calciner (less wastage and increased thermal efficiency). This would have been expected to improve overall fuel efficiency in the pyroprocessing system, because more clinker could be produced using the same amount of fuel.

See also, id. at p. 5-2 to 5-3.

⁶² Hofmann Expert Report, p. 1-7, Opinion 22 (“...CEMEX could have reasonably concluded that this project did not result in any NOx increases due to improved thermal efficiency and kiln stability.”), p. 5-30.

⁶³ Young Expert Report, p. 7, Opinion 7. *See also, id.* at p. 5 – 6, Opinion 1; p. 7, Opinion 6.

⁶⁴ Young Expert Report, Sec. 4.1.1., p. 35 – 36.

boost fan⁶⁵ improved efficiency, reducing NOx emissions. Dr. Davis makes a similar argument, but does not call it efficiency improvement.⁶⁶

As noted in Section II.E, an improvement in efficiency can be realized in three ways: (1) the same amount of clinker can be made by burning less fuel, which would reduce NOx emissions in lb/hr; (2) more clinker could be made using the same fuel; or (3) more clinker can be made by burning more fuel, up to the capacity of the system, which would increase NOx emissions in lb. The produced documents indicate that the third option was chosen. Figure 2 indicates that clinker production was increased. CEMEX's NOx emissions data,⁶⁷ plotted in Mr. Hofmann's Figure 2-2,⁶⁸ and my reanalysis of this data below in Figure 4 confirm that NOx emissions increased. And fuel use data in the produced documents indicate heat input increased from about 2.2 million Btu/yr in 1995 to 2.5 million Btu/hr in 20001, as summarized in Table 1. Thus, CEMEX burned more fuel to make more clinker, increasing NOx.

Table 1
Historic Fuel Use (MMBtu/yr)

Year	Heat Input	Bates
1992	2,006,559	US00000046
1993	2,047,911	US00000046
1994	2,117,014	US00000046
1995	2,189,457	US00000046
1996	2,255,150	US00000046
1997	2,475,217	CMXLYO00150080
1998	2,695,212	CMXLYO00150087
1999	2,656,302	CMXLYO00150524
2000	2,742,683	CMXLYO00117455
2001	2,454,056	CMXLYO00117455

⁶⁵ Young Expert Report, p. 45 ("Thus, by improving the thermal efficiency and stability of the kiln system, the overall NOx emissions from the Lyons facility were likely reduced by the addition of the annulus fan while some production gains may have been realized.").

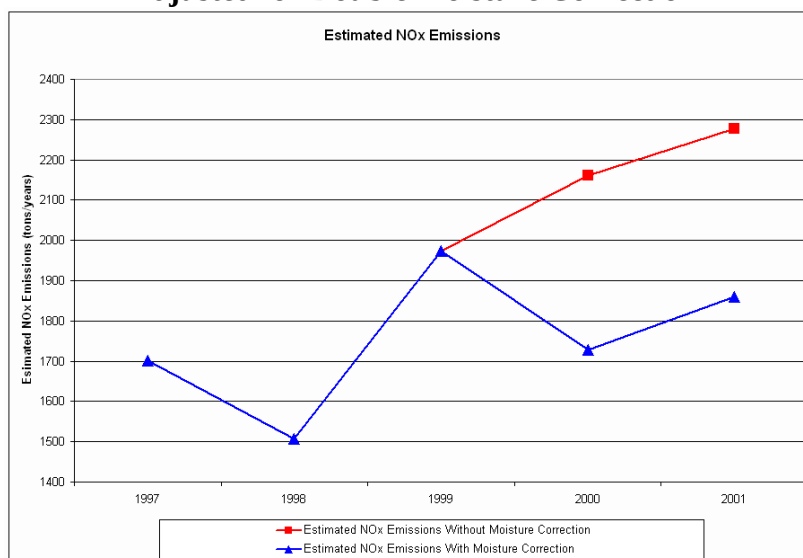
⁶⁶ Davis Expert Report, p. 2 – 3, Opinions 1 – 3, 6.

⁶⁷ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

⁶⁸ Hofmann Expert Report, p. 2-9, Figure 2-2.

The NOx increase was even more significant than suggested by Mr. Hofmann's Figure 2-2. Mr. Hofmann plotted annual NOx emissions as compiled by CEMEX for this case. This compilation and Mr. Hofmann's summary of it underestimate the increase in NOx emissions during the subject period for two reasons. First, the CEMEX data includes two sets of values for 2000 to 2006: estimated NOx emissions without moisture correction and with moisture correction.⁶⁹ Mr. Hofmann plotted the moisture-corrected data, which was erroneously double corrected for moisture in 2000 and 2001,⁷⁰ thereby underestimating NOx emissions in those years.⁷¹ An extract of Mr. Hofmann's data is shown in Figure 3. The blue line (triangles) plots Mr. Hofmann's data. The red line (squares) plots the NOx data without the double moisture correction for 2000 to 2001. When the correct data is plotted, an even steeper increase in NOx is apparent.

Figure 3
Hofmann NOx Emissions Data
Adjusted for Double Moisture Correction



Second, the 1998 and 1999 NOx emission estimates used in Mr. Hofmann's Figure 2-2⁷² exclude periods when the kiln was operating but the CEMS was not. Using the hourly NOx

⁶⁹ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

⁷⁰ Westlin Rebuttal Report, p. 16 (from January 2000 up to October 2001, or 22 months).

⁷¹ See, e.g., CEMEX plot of NOx emission data at Chart2, CMXLYO00038900, which shows CEMEX use of properly corrected data.

⁷² CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845; CEMEX Lyons Facility, Estimated Clinker Production – Historical Summary, CMXLYO00628849.

CEMS data produced by CEMEX for 1998 and 1999 (Exhibit 1), I adjusted CEMEX's estimates of monthly and annual NOx data⁷³ to include NOx emissions for the hours when the kiln was operating, but the CEMS was not collecting data. This analysis is presented in Exhibit 2. In 1998, the CEMS did not collect any NOx data during 967 hours when the kiln was operating. From January to October 1999,⁷⁴ the CEMS did not collect any NOx data during 1,102 hours when the kiln was operating. I was unable to make any estimate of missing hours for the last two months of 1999 as CEMEX did not produce the hourly NOx CEMS data. Thus, the annual NOx emissions in these years have been underestimated by CEMEX.⁷⁵

My analysis in Exhibit 2 indicates NOx emissions of **1,803 tons** in 1998 (compared to 1,507 tons estimated by CEMEX) and **2,232 tons** in 1999 (compared to 1,974 tons estimated by CEMEX). These data are plotted on Figure 4 together with the moisture-corrected data discussed above. This figure shows a very significant increase in NOx emissions over the period 1997 to 2001. This increase in NOx emissions is consistent with CEMEX's internal estimates, which show an increase from 1,714 ton/yr in 1997 to 2,364 ton/yr in 2001,⁷⁶ or a total increase over the expansion period of 650 ton/yr.

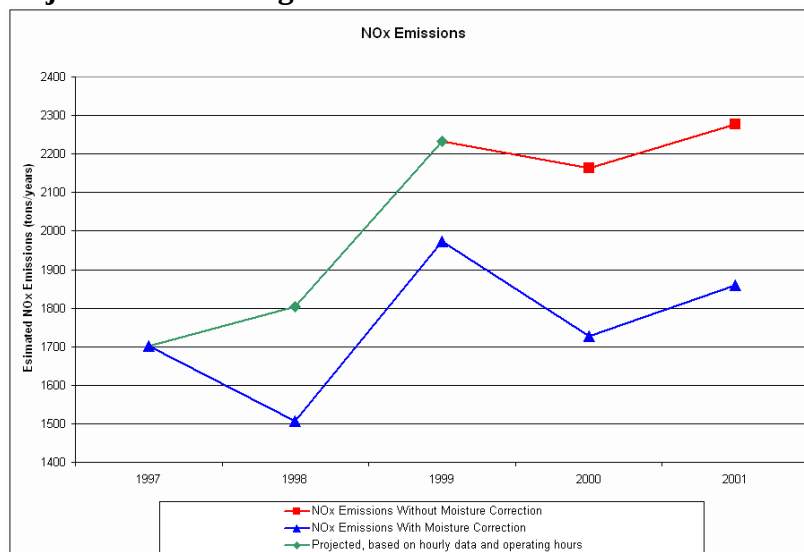
⁷³ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO000628845; CEMEX Lyons Facility, Estimated Clinker Production – Historical Summary, CMXLYO000628849.

⁷⁴ The data acquisition system (DAS) component of the CEMS in use until late 1999 was developed by Odessa Engineering. Sometime in November 1999, CEMEX installed and operated a new data acquisition system (DAS) produced by WTC Engineering. See Deposition of Gregory Wright, November 9, 2010, TR at 15:6 – 14 (“November of ’99, in fact.”); Deposition of Monica Sowders, January 5, 2011, TR at 86:10 – 14 and 86:19 – 22; and Deposition of Chuck Kendrick, January 11, 2011, TR at 8:19 – 24.

⁷⁵ CEMEX did not perform data substitution to account for these gaps until required to do so under its Title V permit in 2000.

⁷⁶ Chart 2, CMXLYO00038900. See also, CEMEX, Environmental Strategic Plan, Lyons Plant, May 2007, CMXLYO000151454 at 151466 (slide 13) and 151485 (slide 32); NOx Trend (lbs/hr), CMXLYO00038900 at 38902, which shows an increase from 442 lb/hr in 1997 to 597 lb/hr in 2001, or a total annual increase of $(597 - 442) \times 8760 \times 0.9 / 2000 = 540$ ton/yr, assuming 90% uptime.

Figure 4
Hofmann NO_x Emissions Data
Adjusted for Missing Data and Double Moisture Correction



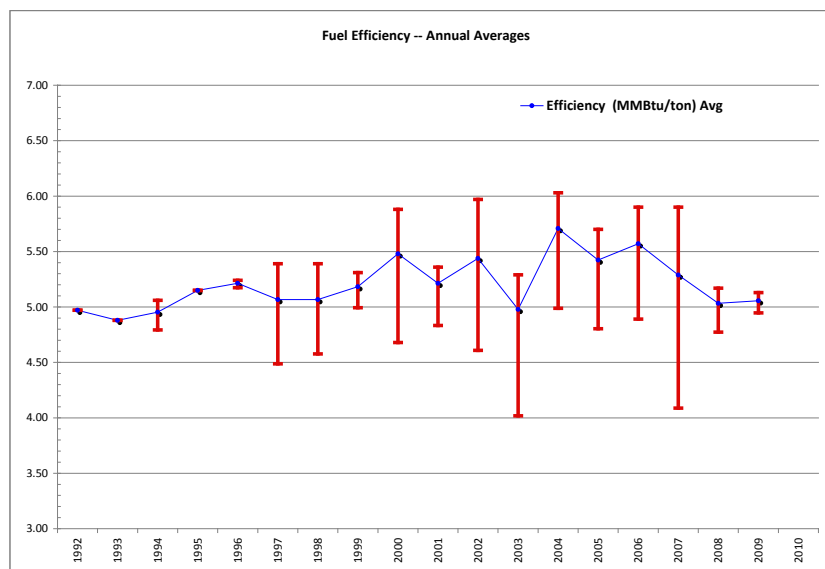
CEMEX has argued that the subject changes allowed the plant to operate more efficiently, requiring less fuel to produce the same amount of clinker. However, CEMEX's actual clinker (Figure 2) and NO_x data (Figure 4, blue line with triangles), even before adjustment for errors, indicate that these projects were not implemented to burn less fuel and reduce NO_x but rather to increase clinker production and increase NO_x.

To investigate CEMEX's efficiency claim, I compiled both annual and monthly fuel efficiency data from the records produced by CEMEX. The data I used are included in Exhibit 3. Fuel efficiency data are recorded in a number of internal reports, including analyses of manufacturing costs, daily and monthly operations summaries, monthly financial reports, fourth day commentaries, and profit analyses.

The data are summarized in Figure 5. The plotted points (triangles) are the average and the error bars (red vertical lines capped on top and bottom) are the range of the reported monthly data. The line follows the mean of the measurements for each year. The efficiency for many time periods varies from report to report. Fuel efficiency is not measured directly at Lyons but rather calculated from the quantity of fuel burned, infrequently measured heat content of the fuel, and the quantity of clinker produced.⁷⁷ This figure shows that the overall fuel efficiency of the pyroprocessing system generally got worse (grew larger) from 1992 (4.97 MMBtu/ton) until 2006 (5.62 MMBtu/ton), when it started to decline (improve). Contrary to CEMEX's contentions, it did not significantly improve (decline) between 1997 and 2001, the period when CEMEX implemented the subject changes.

⁷⁷ Email from RD Smith to Naseem Khan, Re: Fuel Efficiency, November 14, 2003, CMXLYO00045243.

Figure 5
Annual Average Fuel Efficiency



It appears that CEMEX itself did not expect the fuel efficiency to decline. CEMEX used “target” or “plan” fuel efficiencies as guides in planning each year and each month’s work. These represent CEMEX’s goals. The planned fuel efficiencies for 1995 to 2003 are summarized in Table 2. See also the monthly planned fuel efficiencies in Exhibit 3. The data in this table show that CEMEX did not expect the fuel efficiency to decline, as claimed by CEMEX’s experts after the fact.

Table 2
Plan Fuel Efficiency⁷⁸ (MMBtu/ton)

Year	Plan
1995	5.00
1996	5.10
1997	5.10
1998	5.10
1999	5.00
2000	5.00
2001	5.00
2002	5.44
2003	5.71

The fuel efficiency plan in 1995, before any of the subject changes was conceived, was 5.0 MMBtu/ton.⁷⁹ This increased to 5.10 MMBtu/ton in 1996 (got worse) and remained at this level from 1997 to 1998, following the calciner improvements. Thereafter, CEMEX did not plan for an improvement in efficiency from the calciner modifications, *i.e.*, no change from 1997 to 1998. See Table 2 and Exhibit 3. From 1999 to 2001, planned fuel efficiency returned to 5.0 MMBtu/ton, the pre-modification level, and thereafter it got much worse.

Mr. Hofmann only attempted to quantify the efficiency improvement for one of the projects, the calciner improvement. Mr. Hofmann argues the 1997 calciner change improved efficiency from 5.15 MMBtu/ton in 1995 and 5.25 MMBtu/ton in 1996 to 4.81 MMBtu/ton in 1997, showing an 8% improvement in fuel efficiency.⁸⁰ Mr. Hofmann relied on a planning document used for permitting that disagrees with internal efficiency numbers CEMEX reported to management.

The monthly efficiency data for the 1995 to 1997 period selected by Mr. Hofmann is summarized in Figure 6. The blue bars plot monthly efficiency reported by CEMEX. Exhibit 3. The red bar, for March 1997, is the month when the calciner improvement was made. These data (Exhibit 3) indicate that the average efficiency in 1995 to 1996 was 5.34 MMBtu/ton, which declined to 5.21 MMBtu/ton in 1997, or a 2% improvement. This is within the error of the estimate. Subsequent data in Figure 6 indicate this drop was not sustained. There was no improvement in the month immediately following the calciner improvement, April 1997; fuel

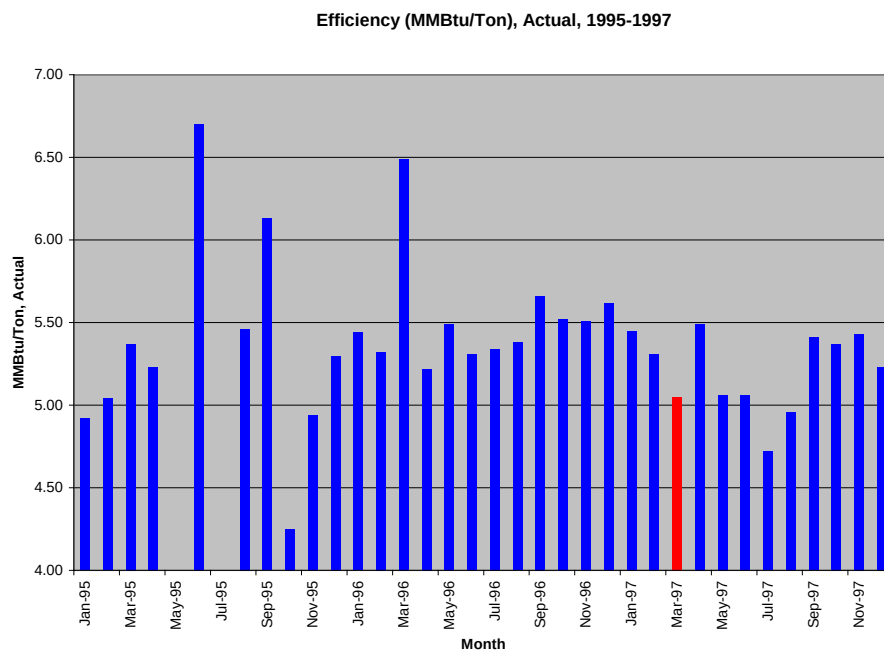
⁷⁸ CEMEX, Lyons Plant, December 2002 MTD, CMXLYO00045217 at 45218; Lyons Kiln History and Forecast (1992 – 1997), CMXLYO000155750; Lyons Plant – Monthly Report of Operations, CMXLYO00043081 at 43083 (December 1999), CMXLYO00043386 at 43388 (January 1999), CMXLYO00047047 at 47086 (December 1995), CMXLYO00043894 at 43895 (May 2001); Untitled Spreadsheet (2000 – 2007), CMXLYO000117455; Variance Analysis, CMXLYO00045236 (July 2003), CMXLYO00045237 (August 2003), CMXLYO00045245 (November 2003).

⁷⁹ Memorandum from John Lohr to Alan Redeker, July 21, 1998, Re: Potential Permit Impacts on Plant Expansion Project, CEMEX_NEIC_00003421 at 3422 (“Historic Level is 5.0 MMBTU/Clinker Ton”).

⁸⁰ Hofmann Expert Report, p. 5-3.

efficiency got worse, going from 5.05 MMBtu/ton in March 1997 to 5.49 MMBtu/ton in April 1997. Oxygen injection started the next month (May 1997),⁸¹ thereby obscuring any further conclusions as to the calciner improvement taken alone.

Figure 6



IV. AGGREGATION

Aggregation is a term of art that refers to the grouping of projects for the purpose of determining PSD applicability. The outcome of a PSD applicability determination depends on what pollutant-emitting activities are considered to be part of a project. Preconstruction review is sometimes evaded by breaking a large project into smaller pieces that do not individually trigger NSR review.⁸² As none of the subject projects was permitted (the oxygen plant is “mentioned” in the Title V permit but never received a preconstruction permit), in my Expert Report, I evaluated the subject projects taken both collectively, *i.e.*, aggregated, and individually to determine if a PSD permit should have been obtained prior to construction.⁸³ Mr. Hofmann

⁸¹ Fox Expert Report, p. 5, Table 1; Deposition of Kevin Kelley, December 16, 2010, TR at 154:2 – 154:5 and 154:20 – 154:22.

⁸² NSR Manual, p. A.36 (“A deliberate decision to split an otherwise ‘significant’ project into two or more smaller projects to avoid PSD review would be viewed as circumvention and would subject the entire project to enforcement action if construction on any of the small projects commences without a valid PSD permit.”).

⁸³ Fox Expert Report, p. 9 – 29.

argues that the subject projects should not be evaluated collectively or “aggregated.” Instead, he argues that some of these projects – the calciner combustion project, the ID fan motor replacement, and the annulus boost fan – should be analyzed individually because they were independent projects.⁸⁴ Mr. Hofmann’s conclusions are incorrect because he has ignored the record in this case.

A. CEMEX’s Expansion Plan Was a Response to Market Conditions

The Colorado cement market had been oversold since 1992 and presented an attractive business opportunity for CEMEX.⁸⁵ The documents produced by CEMEX indicate that in response to these market conditions, CEMEX had begun a concerted effort to increase cement production, starting at least as early as 1996, when it was investigating the calciner improvement, the finish mill, a new separator,⁸⁶ and oxygen injection.⁸⁷ It likely began before then, based on the clinker production data summarized in Figure 2. However, the produced documents do not cover this earlier period. All of these changes were subsequently incorporated into the project Mr. Hofmann refers to as the Lyons Expansion Project, or the “LEP.”⁸⁸

⁸⁴ Hofmann Expert Report, p. 1-4, Opinion 8; p. 5-1 (“[M]y opinion is that not all of these projects should be considered part of the Lyons Expansion Project and would not have been properly aggregated for PSD purposes.”); p. 5-22 (“[A]ll of these projects would have been implemented as stand-alone projects due to the independent manufacturing, quality aspects, and fuel use/cost benefits associated with each. There is no indication that CEMEX senior management required Mr. Lohr to justify these projects as a group or as part of an overall plant expansion plan. In fact, there is ample evidence these projects were independently conceived and implemented.”).

⁸⁵ Memorandum from Karen Krenek to Norris Widener, November 18, 1998, Re: Approved AFE, Lyons Capacity Increase, CEMEX_NEIC_00003449 – 3459 at 3453.

The demand for cement in Colorado has exceeded the state's capacity since 1992. In 1996, total cement demand amounted to 2.1 million tons, compared to in-state capacity of 1.8 million-tons. Demand is projected to exceed 2.3 million tons in 1997. . . . There is little doubt that the Lyons plant could have sold an additional 100,000 to 125,000 tons of cement in the Colorado market during 1997 if the product had been available.

Id. at CEMEX_NEIC_00003453.

⁸⁶ Technical Services Group, Activity Report, September 1, 1996, US00078597 at 78767 (“3. Lyons ... (4) Investigated finish mill and separator operation and designed first compartment ball charge with resulting in increase in production rate and cement quality.... (6) Worked with FCT on review of combustion problems...”).

⁸⁷ Herman Tseng & John W. Lohr, Oxygen Enrichment to Calciner Firing, CMXLYO00612639 (Goodrich Deposition Exhibit 365) (“Between 1995 and 1999, the US cement market grew at an average of 7%. In response, domestic manufacturers looked at all available means to increase plant capacity. During 1996, Southdown reconsidered the potential of oxygen enrichment...”). This appears to be a draft of the paper entitled “Oxygen Enrichment” that was published in International Cement Review in May 2001, CMXLYO00612451.

⁸⁸ This project was variously called the “Lyons Capacity Increase” (Memorandum from Karen Krenek to Norris Widener, November 18, 1998, Re: Approved AFE, US00000454 at 454, 455); “Lyons Plant Expansion” (Memorandum from Karen Krenek to Norris Widener, November 18, 1998, Re: Approved AFE, US00000454 at 456 457, 462); “Phase I Expansion” (Southdown, Inc., Authorization for Expenditure, November 18, 1998, US00000027 at 37 – 38; Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, CEMEX_NEIC_00003247 at 3251); “Plant Capacity Increase” (Letter from David Repasz, Southdown, Inc., to

All of the subject projects had a singular goal, conceived at least as early as mid-1996, which was to increase clinker production to reach a goal of about 600,000 ton/yr of cement. They are thus related, each a step toward the same goal and therefore should be considered together for PSD purposes.⁸⁹ The production increase target could not have been achieved without all of these projects.⁹⁰ CEMEX repeatedly refers to all of these projects together, in numerous CEMEX documents, as improvements designed to increase throughput,⁹¹ indicating CEMEX's intent. This is demonstrated below for each project Mr. Hofmann claims should be separately evaluated.

Kenneth M. Hitcho, Humboldt Wedag, Inc., June 12, 1998, Re: Letter of Intent for Supply to Finish Grinding Separator System for Lyons, Colorado Plant Authorized Spending, CEMEX_NEIC_00003381 at 3382); "Plant Expansion Project," (Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, CEMEX_NEIC_00003247 at 3248); and "Lyons Expansion Project" (Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update, CEMEX_NEIC_00003247 at 3249), among others, in CEMEX internal documents. Mr. Hofmann assigns the moniker of LEP or Lyons Expansion Project.

⁸⁹Memorandum from John B. Rasnic, Director, Stationary Source Compliance Division, Office of Air Quality Planning and Standards, to George T. Czerniak Chief, Air Enforcement Branch, Region V, Re: Applicability of New Source Review Circumvention Guidance to 3M – Maple Minnesota, June 17, 1993. Available at: <http://www.epa.gov/region7/air/nsr/nsrmemos/maplwood.pdf>.

Where a source is permitted for several minor modifications that may in good faith be intended to be separate but result in the source's aggregate increases to be major even considering decreases over a short time period (e.g., one year or 18 months), the modifications may require major new source review. Such modifications could require NSR if they are viewed as being consistent with the source's overall production goals or plans for a short planning period. In other words, 3M should not benefit from the absence of a plant-wide production plan.

Id. at p. 4.

⁹⁰ Mr. Hofmann erroneously attributes clinker production increases to improvements in kiln uptime. Hofmann Expert Report, p. 3-1. However, his own data in Table 5-6, p. 5-25, indicates there was no increase in kiln uptime between 1996 and 2001. Rather, the increases in production were due to specific projects that together removed various process bottlenecks to increase cement production in an oversold market.

⁹¹ PowerPoint, CEMEX, Inc., Lyons Cement Plant, CMXLYO00150942 at 150947, "Kiln Throughput History" (includes in the list combustion chamber modification, ID fan HP increase, raw mix and CKD use, liquid and VSA oxygen, annulus fan, kiln feed screw, alkali return, and low volatile coal); PowerPoint, CEMEX, Inc., Lyons Cement Plant, CMXLYO00151197 at 151201, "Kiln Throughput History" (listing "improvements/increases: combustion chamber modification (+80 t/d), ID fan HP increase (+28 t/d), raw mix & CKD utilization (+85 t/d), liquid and VSA oxygen (+155 t/d), annulus fan, kiln feed screw, alkali bypass dust return (+77 t/d), low volatile coal (+150 t/d), and riser coal firing (+50 t/d)"); Lyons Clinker Production History, May 17, 2000, CMXLYO00045592 (listing projects that increased production, including, in 1997, combustion chamber modification, ID fan horsepower increase, and oxygen enrichment and in 2000, annulus boost fan kiln combustion, among others).

1. Calciner Combustion Project

The calciner is the part of the pyroprocessing system that pretreats or calcines the raw feed.⁹² During the March 1997 kiln outage, the calciner combustion chamber was modified to improve combustion efficiency and increase clinker production.⁹³ I evaluated PSD applicability for this change both individually and collectively, with the other subject changes.⁹⁴

Mr. Hofmann argues that the calciner combustion chamber project should be analyzed individually for PSD purposes because it was an “independent project driven solely by an operational problem well known to the plant management years before the other projects were even conceived, planned, or budgeted.”⁹⁵ Mr. Hofmann asserts this project “was also not part of any of the documents prepared as the Lyons Expansion Project” and thus should not be aggregated with any other projects.⁹⁶ These claims are wrong.

CEMEX itself links the calciner improvement project with its expansion plan. CEMEX was investigating increasing cement production as early as September 1996, at the same time it was studying the calciner project and other LEP projects.⁹⁷ CEMEX’s description of the “Lyons Plant Expansion,” dated January 1998 and included in the approved Authorization for Expenditure (AFE) for the expansion (LEP), links the calciner improvement project with the capacity expansion: “A portion of this capacity increase was achieved in 1997 through minor modifications to the preheater aerodynamics combined with the temporary use of high cost purchased oxygen.”⁹⁸ This indicates the calciner improvement was required to achieve

⁹² Deposition of Andrew Jackura, September 16, 2010, TR at 26:12-16 (explaining what a combustion chamber is: “On that kiln, the combustion chamber is the secondary firing vessel in the preheater. It’s a place other kilns might call a calciner.”); Deposition of Steven Mossberg, September 8, 2010, TR at 27:2-7 (“For clarification, we had a two stage preheater, and one part of the first chamber was – the whole unit would be the calciner, but the first part was the combustion chamber, and then there was a...calciner.”).

⁹³ Lyons Clinker Production History, May 17, 2000, CMXLYO00045592; Case Study of Physical Modeling at Southdown’s Lyons Plant, CMXLYO00402635 – 402640; Monthly Financial Report, March 31, 1997, CMXLYO00049696 at 49753; Monthly Financial Report, April 30, 1997, CMXLYO00049893 at 49947; Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, January 4, 1998, Re: Lyons Kiln Production and Oxygen Use, CMXLYO00039059 at 39060 (the combustion chamber modifications would increase clinker production from the historic level of 54.5 ton/yr to 60.0 ton/hour without oxygen, 62.0 tons/hour with 16 tons/day oxygen, and 65.6 tons/hour with 45 tons/day oxygen).

⁹⁴ Fox Expert Report, Table 3 (changes taken collectively), Table 5, Item #1 (calciner modifications taken individually).

⁹⁵ Hofmann Expert Report, p. 1-4, Opinion 9.

⁹⁶ Hofmann Expert Report, p. 5-3.

⁹⁷ Technical Services Group, Activity Report, September 1, 1996, US00078597 at 78767.

⁹⁸ Memorandum from Norris Widener to Karen Krenck, Re: Approved AFE, November 18, 1998, US00000454 at 457.

CEMEX's production goal and specifically links it with oxygen injection, which CEMEX's experts acknowledge was part of the Lyons Capacity Increase.⁹⁹

In October 2000, retrospectively, CEMEX further explained: "We made conscious changes in raw mill operation beginning early in 1997 with the modification of the calciner combustion chamber. Immediately after this change we purchased a higher horsepower motor for the ID fan. These two changes combined produced about a 10% increase in kiln throughput."¹⁰⁰ Thus, CEMEX links the calciner project with the raw mill project, which Mr. Hofmann classifies as part of the Lyons Expansion Project.¹⁰¹

Looking back over the history of its expansion, CEMEX repeatedly classifies the calciner project as a capacity increase project along with others Mr. Hofmann aggregates as the LEP.¹⁰² Thus, the increase in capacity anticipated by the LEP depended upon the earlier calciner project, which was specifically linked to it by CEMEX itself, was studied at a time CEMEX was generally investigating methods to increase cement production due to the oversold Colorado market, and was required to meet production goals CEMEX had set for itself.

2. Induced Draft Fan Motor Replacement

The Induced Draft ("ID") fan supplies air to the kiln. On August 8, 1997, the 1250 hp motor on the ID fan burned out and was replaced with a 1500 hp motor.¹⁰³ I evaluated PSD applicability for this change both individually and collectively, with the other subject changes.¹⁰⁴ Mr. Hofmann argues that the ID fan motor replacement should not be aggregated with the LEP because it was driven by an emergency situation and was not related to the other projects. He argues that "[t]here is no evidence that CEMEX included this replacement in its planning process prior to the event. Capital spending decisions made subsequent to the event do not support reliance on the ID fan motor replacement with respect to any of the other alleged projects."¹⁰⁵

⁹⁹ See Expert Report of Gerald L. Young, May 10, 2011, p. 26 ("At the Lyons facility a problematic condition used to occur in which fuel fired into the calciner does not fully combust. . . This condition is sometimes referred to as 'overburn' and was the reason why the internal geometry of the calciner was modified in 1997. It is also part of the reason why CEMEX decided to inject oxygen into the calciner.") (emphasis added).

¹⁰⁰ Email from John Lohr to Dave Repasz, et. al, September 19, 2000, Re: Power Review.xls, CMXLYO00045599 at 45600.

¹⁰¹ Hofmann Expert Report, p. 5-23, Table 5-5 and p. 5-3.

¹⁰² Email from John Lohr to Dave Repasz, et. al, September 19, 2000, Re: Power Review.xls, CMXLYO00045599 at 45600.

¹⁰³ Hofmann Expert Report, p. 5-6.

¹⁰⁴ Fox Expert Report, p. 20, Table 3 (changes taken collectively) and p. 24, Table 5, Item #4 (ID Fan).

¹⁰⁵ Hofmann Expert Report, p. 1-5, Opinion 11 and Sec. 5.2, p. 5-6.

He further argues this project should not be aggregated because the fan and fan curve did not change.¹⁰⁶

Prior to its replacement, CEMEX was operating the ID fan at maximum capacity.¹⁰⁷ Historically, CEMEX characterized the kiln as being limited by the ID fan.¹⁰⁸ CEMEX also knew that the ID fan was at the end of its useful life.¹⁰⁹ The capital projects list for 1992 to 1999 describes the replacement of the ID fan motor as the “kiln ID fan motor upgrade.”¹¹⁰ Viewed as a whole, the record suggests that CEMEX ran the ID fan motor until it failed, thus deferring the replacement expense until it could no longer be avoided. CEMEX also knew that the larger motor it had installed, when run at a higher speed, might trigger PSD review.¹¹¹ In fact, oxygen was used in an effort to avoid increasing the size of the ID fan.¹¹²

Mr. Hofmann also opines that “it would have been reasonable for CEMEX to assume that no extra RPMs/will [sic] be gained from replacement” because the fan and fan curve were not changed.¹¹³ Mr. Young, on the other hand, admits the larger motor would increase emissions,

¹⁰⁶ Hofmann Expert Report, p. 5-7.

¹⁰⁷ Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00402688 at 402692 – 402693 (“amped out on ID Fan (1029-1031 RPM)”); See Deposition of John Lohr, October 11, 2010, TR at 60:4-17 and 61:17-23 as to “amped out.” Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, January 4, 1998, Re: Lyons Kiln Production and Oxygen Use, CMXLYO00039059 at 39060 (“The kiln has been our bottleneck for most of the plant’s history. We have had raw feed production and finish mill capacity available.”).

¹⁰⁸ Southdown, Inc. Lyons Plant, Process Additions, Steel Slag and Oxygen Injection, March 7, 2000, CMXLYO00045630 at 45631 (“Plant is Quarry and ID Fan Limited. Soldout market since 1991.”) and 45639 (“ID Fan Limited on Kiln Combustion.”); John Lohr, Plant Production Bottlenecks, Lyons, Colorado, January 14, 2004, CMXLYO00044998 at 44999 (“The kiln system capacity is typically limited by draft. The plant runs the ID fan at maximum on a continuous basis.... Permit modifications would likely be required to increase the ID fan size or to make major modifications to ductwork.”) (emphasis added).

¹⁰⁹ Hofmann Expert Report, p. 5-7.

¹¹⁰ CEMEX, Inc., Lyons Cement Plant, Capital Projects 1992 to 1999, CMXLYO00180114 at 180115, Project #76, Start Date 8/97, End Date 12/97.

¹¹¹ Southdown, Inc. Lyons Plant, Process Additions, Steel Slag and Oxygen Injection, March 7, 2000, CMXLYO00045569 at 45578 (The historic situation is ID fan limited and “[a]ny fan change would trigger PSD.”).

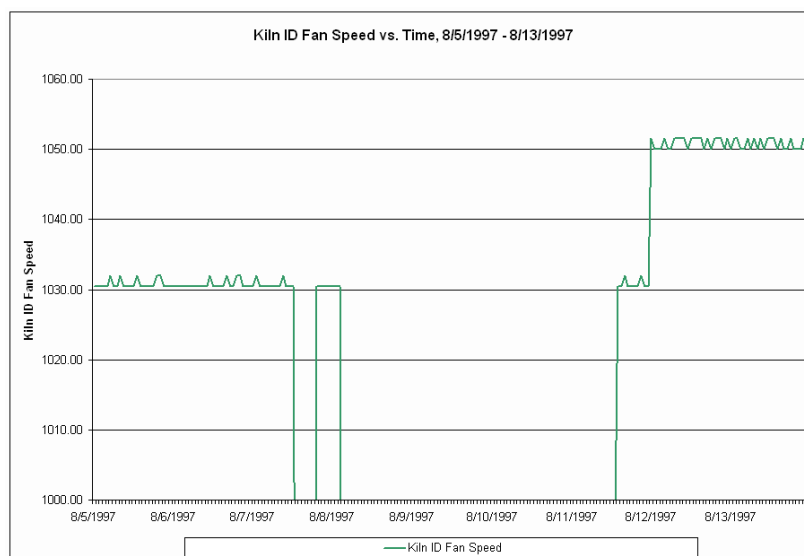
¹¹² Herman Tseng & John W. Lohr, Oxygen Enrichment to Calciner Firing, CMXLYO00612639 (Goodrich Exhibit 365) (Prior to the oxygen injection tests: “It was appreciated that, in many cases, a more economic approach would be to upgrade the kiln ID fan. However, the capital cost, permit requirements, equipment lead time, and kiln downtime for installation would prevent rapid response to an immediate need for increased production....”). This appears to be a draft of the paper entitled “Oxygen Enrichment” that was published in International Cement Review in May 2001, CMXLYO00612451. Southdown, Inc. Lyons Plant, Process Additions, Steel Slag and Oxygen Injection, March 7, 2000, CMXLYO00045569 at 45578 (The historic situation is ID fan limited and “[a]ny fan change would trigger PSD.”). John Lohr, Plant Production Bottlenecks, Lyons, Colorado, January 14, 2004, CMXLYO00044998 at 44999 (“Permit modifications would likely be required to increase the ID fan or to make major modifications to ductwork.”).

¹¹³ Hofmann Expert Report, p. 5-7.

but argues it would not be in direct proportion to the increase in fan speed due to air leakage and would be too small to notice. He opines that the original ID fan motor had limited the fan speed to 1030 rpm and, after the motor was replaced, the fan was “eventually” operated at 1070 rpm (a 3.88% increase). He then speculates, without citing any record evidence, that this would not allow 3.88% additional fuel to be burned due to air leakage but rather only an increase of 1.5% to 2.5%. He further speculates that such a small increase would not have alerted Lyons staff that it might significantly increase NOx.¹¹⁴ This is incorrect.

The daily Operations Reports show that CEMEX took advantage of the larger motor, immediately ramping up the fan speed from 1030 to 1050 rpm on startup. This is illustrated in Figure 7, which plots the hourly data from August 5, 1997 to August 13, 1997.¹¹⁵ This figure shows that on startup with the new ID fan motor, the ID fan was operated at 1030 rpm for 10 hours and then ramped up to 1050 rpm.

Figure 7



The fan speed data for 1997 to 1998 are plotted in Figure 8 from the daily Operations Reports.¹¹⁶ These data indicate the new motor was continuously operated at higher speeds than before the failure, increasing from 1050 rpm to 1080 rpm.¹¹⁷ Most of the data indicates that the ID fan motor routinely ran at 1060 rpm to 1080 rpm. The motor was apparently limited by fan vibration issues¹¹⁸ and slated for repair in 2000.¹¹⁹

¹¹⁴ Young Expert Report, p. 37 – 38.

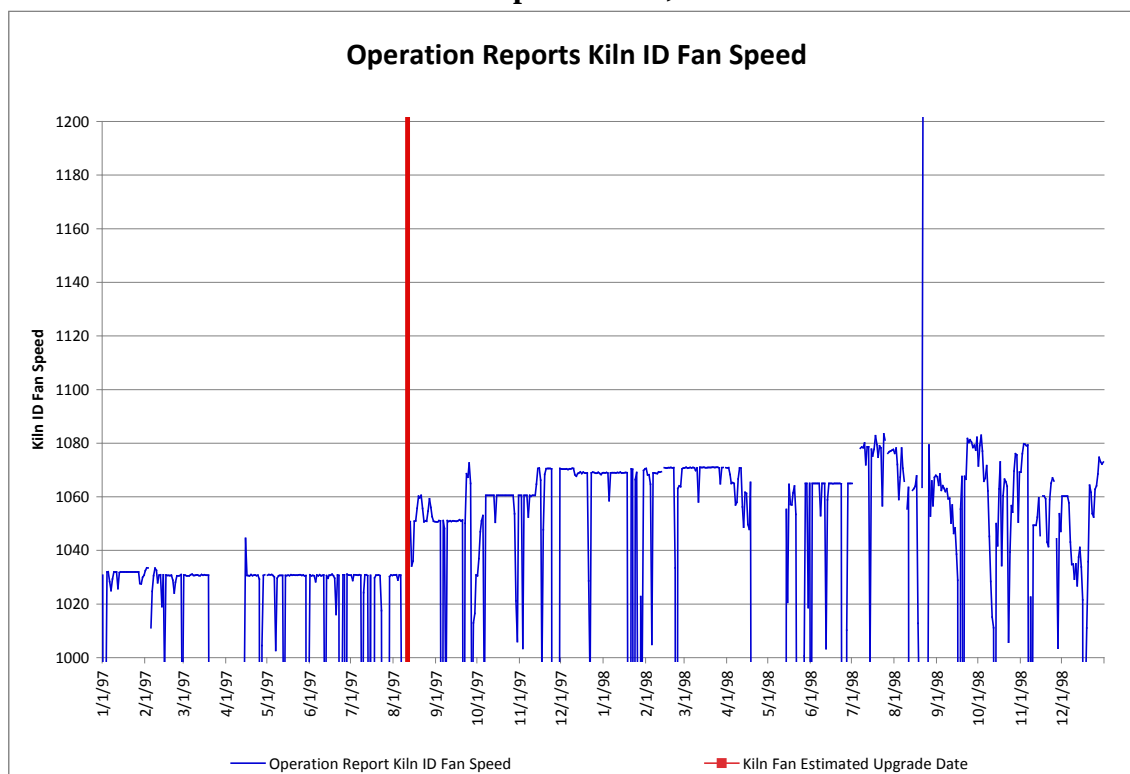
¹¹⁵ Plotted data from Daily Operations Reports at CMXLYO00087524 – 87556.

¹¹⁶ Plotted data from Daily Operations Reports at CMXLYO00085234 – 95452.

¹¹⁷ Daily Operations Report, August 22, 1998, CMXLYO00086283 at 86284.

¹¹⁸ Daily Production Notes, September 21, 1998, CMXLYO00084921 and Lyons Plant Cost Reduction Meeting, March 18 & 19, 1999, CMXLYO00045695 at 45697.

Figure 8
Kiln Drive Amps vs. Time, 1998 – 1999



Although the fan and fan curve did not change, the fan speed was immediately increased after the new motor was installed, which CEMEX knew would increase kiln production.¹²⁰ Thus, while the specific timing for installation of the larger motor was determined by an emergency, the point at which CEMEX decided to take advantage of the larger motor to increase production, 10 hours after startup, should have caused CEMEX to approach the Colorado Department of Public Health and Environment (CDPHE) to seek either an applicability determination, a permit operating limitation on the ID fan, or to submit a application for a PSD permit. CEMEX clearly was aware that changes to the ID fan were likely to trigger PSD review.¹²¹ As CEMEX used the bigger ID fan motor for the same purpose as all of the other subject projects, to increase clinker production, the larger ID fan motor and the utilization of the larger motor capacity should be aggregated with other capacity increase projects.

¹¹⁹ Lyons Kiln Slowdowns in 1999 (June-December), CMXLYO00157286.

¹²⁰ See Rebuttal Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011 (hereinafter "Greer Rebuttal Report").

¹²¹ Lyons Clinker Production History, May 17, 2000, CMXLYO00045592 (ID fan horsepower increase = 12,483 ton/yr); Riser Coal Firing, Lyons, Colorado, April 9, 2002, CMXLYO00045284 at 45288 ("ID fan HP increase +28 ton/day"); Southdown, Inc. Lyons Plant, Process Additions, Steel Slag and Oxygen Injection, March 7, 2000, CMXLYO00045569 at 45578 (The historic situation is ID fan limited and "[a]ny fan change would trigger PSD."). See Greer Rebuttal Report.

3. Annulus Boost Fan

The coal tip velocity of the kiln burner was low compared to optimum levels for a direct-fired burner, which resulted in a long and lazy flame, producing a long burning zone.¹²² The solution selected by CEMEX was to install an annulus boost fan to boost the coal tip velocity.

In November 1997, CEMEX included an annulus boost fan (coal booster fan) in the expansion: “Coal Booster Fan: ... This proposal will add a high static booster fan to the coal circuit which will increase the tip velocity to the optimum level.”¹²³ This fan was not installed until May 2000.¹²⁴ I evaluated PSD applicability for this change both individually and collectively, with the other subject changes.¹²⁵

Mr. Hofmann recognized that an annulus boost fan was originally included within the scope of the LEP, but claims, without citation, that it was abandoned in 1998. He also claims, without citation, that an independent decision, distinct from the LEP, was made in 2000 to install a similar fan. Thus, he opines that the annulus boost fan need not be aggregated for PSD purposes.¹²⁶ Further, he argues the timing and nature (to improve combustion characteristics and product quality) of the annulus boost fan project do not satisfy EPA’s aggregation policy guidance with respect to earlier projects.¹²⁷ This is incorrect.

First, the annulus boost fan should be aggregated with other subject projects as it was always officially part of the LEP. The AFE for the annulus boost fan was approved 7/22/98 and carried over as of 8/28/98, scheduled for completion by 3/31/99.¹²⁸ It was still listed in the 2000 budget plan, slated to be completed by 1/0/00.¹²⁹ The project was postponed due to the potential downside risk of miscalculating the correct burner pipe momentum/placement/alignment or not accounting for other significant process issues. It was not altered, however, but rather put on hold until this uncertainty was resolved.¹³⁰ By 1999, CEMEX had decided to engage a

¹²² See description in Lyons Plant, 2000 Plan, January 14, 2000, CMXLYO00415808 at 415814.

¹²³ Memorandum from Gregg St. Clair to Dave Repasz, November 19, 1997, Re: Lyons Plant Expansion – Phase I, CEMEX_NEIC_00003291 at 3296.

¹²⁴ Fox Expert Report, p. 5, Table 1.

¹²⁵ Fox Expert Report, p. 20, Table 3 (changes taken collectively) and p. 24, Table 5, Item #10 (Annulus Boost Fan).

¹²⁶ Hofmann Expert Report, p. 5-30.

¹²⁷ Hofmann Expert Report, p. 5-22.

¹²⁸ Lyons Plant, August 28, 1998, CMXLYO00155783.

¹²⁹ Lyons Plant, 2000 Plan, January 14, 2000, CMXLYO00415808.

¹³⁰ Text and Figures for FCT AFE, CMXLYO00045595.

consultant to complete an audit and system model to determine the needed process changes.¹³¹ Thus, it was and remained part of the LEP, so under Mr. Hofmann's reasoning, should be included with other LEP projects for determining PSD applicability.

Second, as to timing, Mr. Hofmann claims the annulus boost fan project "occurred nearly three years after the projects primarily associated the Lyons Expansion Project [sic]."¹³² This is incorrect, as demonstrated by the project chronology in my expert report.¹³³ The annulus boost fan was installed in August 2000,¹³⁴ a little over a year after the VSA oxygen plant, which was viewed as the heart of the expansion. This is within the timeframe allowed under the aggregation guidance cited by Mr. Hofmann.¹³⁵

Third, the annulus boost fan should be aggregated with other subject projects because its primary goal was to increase production. Mr. Hofmann concedes a "modest increase in clinker production."¹³⁶ In the AFE to hire FCT to design the annulus boost fan, CEMEX explained:

The Lyons kiln is currently limited by permit to a fixed amount of fuel and feed. Any improvements in efficiency could be expected to be translated into annual capacity up to a maximum of 600,000 cement tons per year (570,000 clinker tons). Current efforts with oxygen injection and mix optimization are expected to push system capacity to about 535,000 clinker or 565,000 cement leaving room for up to 35,000 tons of cement capacity [for the annulus boost fan].¹³⁷

Mr. Hofmann's contention is at odds with CEMEX's contemporaneous expectations for the annulus boost fan.

Preliminary estimates by FCT indicated the annulus boost fan could increase efficiency by 0.4 to 0.5 MMBtu/ton and kiln capacity by 3% to 5%.¹³⁸ The benefits of this project are also described elsewhere as: "Higher tip velocities will produce a hotter, shorter, more stable coal flame. This will, in turn promote better coating and higher production. Kosmosdale reported a

¹³¹ Text and Figures for FCT AFE, CMXLYO00045595; Lyons Plant Cost Reduction Meeting, March 18 & 19, 1999, CMXLYO00045695 at 45696 ("Installation of Coal Mill Booster Fan...Lyons Staff will discuss this project with FCT next week....").

¹³² Hofmann Expert Report, p. 5-22.

¹³³ Fox Expert Report, p. 5, Table 1.

¹³⁴ *Ibid.*

¹³⁵ Hofmann Expert Report, Sec. 4.1.6, p. 4-5. The aggregation guidance cited by Mr. Hofmann never took effect.

¹³⁶ Hofmann Expert Report, p. 5-22.

¹³⁷ Text and Figures for FCT AFE, CMXLYO00045595.

¹³⁸ Text and Figures for FCT AFE, CMXLYO00045595; Lyons Plant Cost Reduction Meeting, March 18 & 19, 1999, CMXLYO00045695 at 45696 ("Installation of Coal Mill Booster Fan...Lyons Staff will discuss this project with FCT next week....").

substantial [sic] (10%) production increase with the installation of their booster fan.”¹³⁹ A hotter flame means more NOx. Higher NOx was observed.¹⁴⁰ Thus, CEMEX itself viewed this project as another method to boost capacity to reach its production goal of 600,000 ton/yr of cement.

V. INCREASE IN EMISSIONS

Mr. Hofmann calculates what he calls a “project increase” in NOx emissions due to several changes at the facility: calciner improvements (Table 5-1); ID fan motor replacement (Table 5-2); and the Lyons Expansion Project (LEP) (Table 5-7). His calculations compare NOx emissions in the two years after each project is built to the two years before it was built. He does not state this is a PSD applicability determination but implies it by making claims such as “under this methodology, CEMEX could have concluded that PSD was not triggered...”¹⁴¹ Elsewhere in his report, he lays out the process used to make a PSD applicability determination¹⁴² but does not actually apply it to the projects at hand, instead just calculating the “project increase,” which as I show below is not a PSD applicability determination. Taken together, this material is misleading, incorrect, and irrelevant to the question at hand, which is whether the subject projects triggered PSD review.

A. CEMEX’s Approach Is Inconsistent with Controlling Regulations

A PSD applicability determination consists of three steps: (1) determine if the project is a physical change or change in method of operation; (2) determine if the emissions from the project all by itself would result in a significant increase in emissions; and (3) if Step 2 emissions exceed the PSD significance threshold (40 ton/yr for NOx), the net increase in emissions is determined in Step 3.¹⁴³ The net increase in emissions is calculated by subtracting emissions after the project is built from emissions before the project is built (baseline). These calculations are referred to as “netting.” As PSD is a pre-construction requirement, the post-construction emissions must be estimated. The only acceptable method for calculating post-project emissions when an emissions unit has not begun normal operations is potential to emit.¹⁴⁴

¹³⁹ Lyons Plant, 2000 Plan, January 14, 2000, CMXLYO00415808 at 415814.

¹⁴⁰ Email from Mike McCabe to Rick Schnarre and Philip Alsop, Re: Clinker, October 17, 2000, CMXLYO00402668 (“The new burner has definitely created a more intense and consiste [sic] flame which has resulted in a smaller burning zone...There has been an increase in NOx, but I need a couple more days to better quantify what this increase has been.”).

¹⁴¹ Hofmann Expert Report, p. 5-5 (CEMEX believed at time there would be no significant increase in NOx emissions: “[T]here appears to be no PSD issues associated with this project.”), p. 5-27 (discussing Table 5-7, Mr. Hofmann notes EPA changed its regulations to allow “actual-to-projected-actual” methodology, which would have allowed CEMEX to conclude PSD was not triggered.)

¹⁴² Hofmann Expert Report, Sec. 4.1.7, 4.1.8, and 4.1.9, p. 4-6 to 4-7.

¹⁴³ These are equivalent to the six steps Mr. Hofmann sets out in p. 4-8, Table 4-1, in greater detail.

¹⁴⁴ Fox Expert Report, p. 6 – 7.

In discussing the PSD regulations, Mr. Hofmann appropriately notes that “actual emissions,” in turn, is defined as:

(1) the average rate, in tons per year, at which the unit actually emitted the pollutant during a two-year period which precedes the particular date and which is representative of normal source operations; (2) the Administrator may presume that the source-specific allowable emissions for a unit are equivalent to the actual emissions of the unit; or (3) for any emissions unit (other than electric utility unit) which has not “begun normal operations” on the particular date, actual emissions shall equal the potential to emit of the unit on that date (40 CFR 52.21(b)(21)). The US EPA takes the position that no unit that has been physically or operationally changed, regardless of the magnitude, has “begun normal operations” and thus that all modified units must use the actual-to-potential test to determine PSD applicability.¹⁴⁵

On this point, I do not disagree with Mr. Hofmann, which is why I used “potential to emit” to estimate post-construction emissions in Step 2 of the PSD applicability determination in my Expert Report.¹⁴⁶ However, Mr. Hofmann diverges from the PSD regulations in his discussion of regulatory reform in Section 4.1.9 of his report. He opines:

[T]he US EPA revised its NSR regulations in 2002 to allow facilities to utilize an “actual-to-projected-actual” methodology for determining applicability (67 FR 80,186). . . . Under this methodology, CEMEX could have concluded that PSD was not triggered since, for the reasons discussed elsewhere in this report, it had reasonable bases for projecting that actual emissions following and resulting from the projects would not have increased, let alone significantly.¹⁴⁷

This methodology has never applied to the CEMEX projects at issue.

First, EPA changed the definition of “actual” for only Step 1 of the PSD applicability test, the emissions from the project taken by itself, to allow the use of a unit’s maximum actual annual emissions for the 5-year period after the change or a 10-year period after the change if the change involves an increase in the emission unit’s potential to emit or capacity. EPA did not extend this modification to Step 2, the net emission increase step. The “actual to potential” test is still the proper test. EPA has explained:

...EPA did not extend the ATPA [actual-to-potential-actual] test to step two of the applicability test. EPA believes that it remains appropriate to subject projects that will result in a significant emissions increase under step one of the process, and

¹⁴⁵ Hofmann Expert Report, p. 4-6 to 4-7 (emphasis added).

¹⁴⁶ Fox Expert Report, p. 14; p. 20, Table 3; p. 27, Table 8.

¹⁴⁷ Hofmann Expert Report, p. 5-27. *See also*, discussion on p. 4-7.

thus, are more likely to adversely impact air quality, to undergo a more conservative examination using the actual-to-potential methodology under step two of the analysis.... As outlined below, the 2002 rules implemented these policy considerations by retaining the previous netting provisions for determining the significant net emissions increase.¹⁴⁸

Second, even as to the use of “future actual” test in Step 1, Mr. Hofmann himself acknowledges that EPA did not revise its NSR regulations to allow facilities to use the “future actual” test until December 31, 2002, and Colorado did not adopt similar revisions until April 16, 2004. This is well past the period of interest in this matter. Thus, Mr. Hofmann’s “actual” emission calculations in Tables 5-1, 5-2, and 5-7 are irrelevant for determining PSD applicability.

Long before CEMEX began making plant changes in 1996, EPA explained its interpretation of the 1980 PSD regulations. If a unit has not “begun normal operations” then actual pre-change emissions are compared to a prediction of potential emissions (the actual-to-potential test); if a utility unit has “begun normal operations,” then actual pre-change emissions are compared to a less conservative prediction of future emissions (the actual-to-projected-actual comparison, also referred to as an actual-to-future-actual or actual-to-actual test).¹⁴⁹ There is a good reason not to apply an actual-to-projected-actual comparison when, like here, a facility has not begun normal operations. Until normal operations begin, the facility does not have any emissions information to adjust its baseline emission factor. This is likely why in his report Mr. Hofmann does not explain how to adjust the NOx emission factors (Section II) to address alleged reductions due to efficiency improvements.

The PSD program is a pre-construction program, which requires a facility to estimate emissions impacts before a project is built, not after the fact. In 1996, when CEMEX was contemplating the subject modifications, CEMEX had not begun normal operations, and hence, it had no representative data upon which it could rely to rationally adjust the pyroprocessing system’s baseline emission factor. In fact, CEMEX made no attempt at all to determine emission impacts for any modification other than oxygen enrichment. With regard to that modification, as explained below, Section VII, oxygen enrichment had no significant impact on NOx emission factors. Therefore, if CEMEX had attempted to determine whether it triggered PSD by applying a modified emission factor based on the oxygen enrichment test – something it never did – it would have come to the same conclusion that McVehil-Monnet and I did.

¹⁴⁸ Letter from Cheryl L. Newton, Director, Air and Radiation Division, to Keith Baugues, Assistant Commissioner, Officer of Air Quality, Indiana Department of Environmental Management, April 4, 2011, p. 2. Available at; <http://www.epa.gov/region7/air/nsr/nsrmemos/atpanet.pdf>.

¹⁴⁹ 56 Fed. Reg. 27630, p. 27633 & n.10 (June 14, 1991).

B. CEMEX's Proposed Actual-to-Actual Comparison Is Inappropriate

Mr. Hofmann's calculations of "project increase" in Tables 5-1, 5-2, and 5-7 do not consider the fact that an improvement in efficiency always has the potential of increasing production unless there is a physical or federally enforceable limit on throughput.¹⁵⁰ The purpose of the expansion projects was to remove various system bottlenecks. Thus, looking just at the two years after a project is built does not tell the whole story. The system had several bottlenecks that were removed in a series of projects. Comparing the two years after a project with the two years before will not capture the potential increase because it does not reflect debottlenecking throughout the entire system.

C. Project Increases

Mr. Hofmann presents "project increases" for three projects, calculated as the difference between NOx emissions in the two years after each project was constructed and the two years before. These "project increase" calculations (Tables 5-1, 5-2, and 5-7) are not PSD applicability determinations under either the 1980 or the 2002 PSD regulations.

1. Calciner Improvement

Mr. Hofmann estimated the "actual" NOx emissions before and after the calciner improvement in Table 5-1 of his report. He subtracted the NOx emissions during the 24 months before the change (1751 tons: March 1995 to February 1997) from the 24 months after the change (1560 tons: March 1997 to February 1999). This calculation indicates a decrease of 191 tons.¹⁵¹ Thus, he concluded the calciner improvement did not result in any NOx increases.¹⁵² This is incorrect.

Mr. Hofmann relied on NOx data compiled by CEMEX for this case.¹⁵³ As explained in Section III of this report, the NOx emissions in this summary for 1998, half of the 24 months included in Mr. Hofmann's tally, are incomplete and thus underestimate post-calciner NOx emissions. The 1998 NOx emission data does not include any emissions for periods when the kiln was running but the CEMS was not providing accurate data, which occurred about 12% of the time, or for 965 hours. See Exhibit 2. Thus, I revised the post-calciner NOx emissions by substituting a revised 1998 NOx emissions estimate that corrects for the missing data omitted by CEMEX—1803 tons.

¹⁵⁰ See discussion of efficiency improvement, *supra*, Sec. III.

¹⁵¹ Hofmann Expert Report, p. 5-5, Table 5-1.

¹⁵² Hofmann Expert Report, p. 5-5; Davis Expert Report, p. 16.

¹⁵³ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

Mr. Hofmann also included two months in his post-calciner period (March and April 1997) when the kiln was down for extended periods for its annual maintenance outage to make the calciner modification and replace 245 feet of brick.¹⁵⁴ Thus, these two months are not representative of routine operation. I started my two-year post-calciner modification in May 1997 and ran it to April 1999. When these corrections are made, the post-calciner NOx emissions are 1942 tons,¹⁵⁵ as compared to Mr. Hofmann's estimate of 1560 tons. The project increase using this corrected post-calciner estimate is **191 tons** (1942-1751=191). This exceeds the PSD NOx significance threshold of 40 ton/yr by a significant amount.

The full production capability of the calciner modification could not be realized until other changes were made, as the system was bottlenecked at the time of this improvement. The following changes, discussed in Section VI, occurred in rapid succession after the calciner modification: (1) raw mill transfer line (5/97); (2) purchased oxygen injection (5/97); (3) ID fan motor upgrade (8/97); (4) new raw feed compressor (12/97); (6) new high efficiency separator (10/98); and (7) quarry expansion (10/98).¹⁵⁶ These projects facilitated CEMEX in taking full advantage of the calciner improvement.

2. ID Fan Motor Upgrade

Mr. Hofmann estimated the “actual” NOx emissions before and after the upgrade of the ID fan motor in Table 5-2 of his report. He subtracted the NOx emissions during the 24 months before motor failure (1739 tons: August 1995 to July 1997) from the 24 months after the motor upgrade (1723.7 tons: September 1997 to August 1999). This calculation resulted in a NOx decrease of 15.3 tons.¹⁵⁷ Thus, he concluded the ID fan upgrade did not increase NOx emissions. This is incorrect.

Mr. Hofmann relied on NOx data compiled by CEMEX for this case.¹⁵⁸ As explained in Section III, the NOx emissions in CEMEX's summary for 1998 and 1999 are underestimated,¹⁵⁹ because they exclude 12% of the hours in 1998 and 13% of the hours in 1999 when the kiln was operating. See Exhibit 2.

¹⁵⁴ Monthly Report of Operations, March 1997, CMXLYO00049696 at 49752; Monthly Report of Operations, April 1997, CMXLYO00049893 at 49947.

¹⁵⁵ Revised NOx emissions from September 1997 to August 1999: (160.2 + 156.0 + 147.1 + 178.9 + 140.1 + 1803 + 98.6 + 172.3 + 203.5 + 159.4 + 211.7 + 104.0 + 148.4 + 199.9)/2 = **1941.6 tons**.

¹⁵⁶ Fox Expert Report, p. 5, Table 1.

¹⁵⁷ Hofmann Expert Report, p. 5-10, Table 5-2.

¹⁵⁸ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

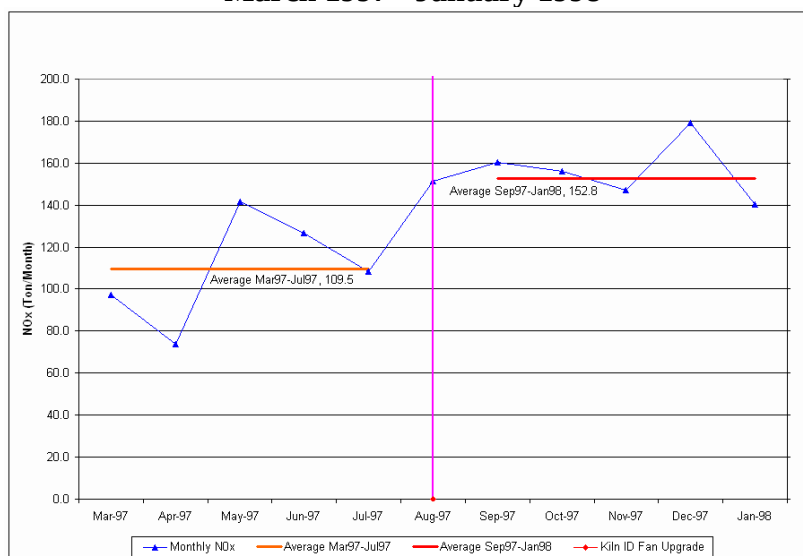
¹⁵⁹ CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

I revised Mr. Hofmann's post-fan NOx emissions estimate by substituting the corrected 1998 emissions estimates (1803 tons) but otherwise used Mr. Hofmann's data. This estimate yields a post-fan NOx emission estimate of 1942 tons,¹⁶⁰ compared to Mr. Hofmann's estimate of 1739 tons. The project increase using this corrected post-fan estimate is **203 tons** (1942-1739=203). This exceeds the PSD significance threshold for NOx of 40 ton/yr by a significant amount.

Further, Mr. Hofmann's two-year look forward is not the best way to isolate the impact of the larger ID fan motor because the new raw feed compressor was installed in December 1997, within the two-year post-fan period, and because the calciner modification, routine oxygen injection, and raw mill transfer line increase preceded it, ratcheting up the baseline. As such, this period does not allow one to isolate the impact of the ID fan motor itself.

Thus, to confirm my estimate of an increase, I isolated the impacts of projects that preceded and followed the ID fan upgrade by comparing the five months prior to the upgrade with the five months after the upgrade. This comparison indicates NOx emissions increased from 109.5 ton/mo to 152.8 ton/mo, or by 43 ton/mo. Assuming 90% runtime, this amounts to an annual increase of 464 ton/yr. This exceeds the PSD significance threshold for NOx of 40 ton/yr by a significant amount. This is not a PSD applicability determination, but rather a demonstration, in rebuttal to Mr. Hofmann, that the upgraded ID fan motor did in fact cause NOx to increase.

Figure 9¹⁶¹
Impact of ID Fan Motor on NOx Emissions (Ton/Mo)
March 1997 - January 1998



¹⁶⁰ Revised NOx emissions from September 1997 to August 1999: (160.2 + 156.0 + 147.1 + 178.9 + 140.1 + 1803 + 98.6 + 172.3 + 203.5 + 159.4 + 211.7 + 104.0 + 148.4 + 199.9)/2 = **1941.6 tons**.

¹⁶¹ NOx Emissions – Kiln – Historical Summary, CMXLYO00628845.

3. Lyons Expansion Project (LEP)

Mr. Hofmann estimated the “actual” NOx emissions before and after the Lyons Expansion Project (LEP) in Table 5-7 of his report. He subtracted the NOx emissions during the 24 months before September 1997 (1753.4 tons) from the 24 months after May 1999 (1787.7 tons). This calculation resulted in an increase in NOx of 34.4 tons.¹⁶² Thus, Mr. Hofmann concluded that CEMEX had a “reasonable bases [sic] for projecting that actual emissions following and resulting from the project would not have increased, let alone significantly.”¹⁶³ This is incorrect.

First, Mr. Hofmann does not provide any explanation for his choice of September 1997 as the LEP start date nor for May 1999 as the end date. Mr. Hofmann’s start date does not make any sense unless you erroneously posit that the calciner and ID fan modification, increased ID fan rpms, and use of liquid oxygen injection are not part of the LEP. The end date employed by Mr. Hofmann is apparently the installation date of the VSA oxygen plant, which ignores the completion of the LEP with the annulus booster fan installation in August 2000.

The start date for Mr. Hofmann’s lookback period should be the 24 months prior to May 1997, as routine oxygen injection, which Mr. Hofmann concedes as part of the LEP, began in May 1997.¹⁶⁴ The 24 months prior to May 1997 yield a baseline of 1694 tons, compared to Mr. Hofmann’s estimate of 1753 tons. This change alone results in an increase in NOx emissions of **94 tons**, which exceeds the NOx significance threshold.

Second, Mr. Hofmann’s post-project estimate is based on double-moisture corrected NOx data for the period January 2000 to May 2001.¹⁶⁵ Using the properly corrected NOx data, captioned “estimated NOx emissions without moisture correction” in CEMEX’s summary, the post-project NOx emissions are 2103 tons, compared to Mr. Hofmann’s estimate of 1788 tons. This change alone results in an increase in NOx emissions associated with the LEP of **350 tons** (2103-1753=350). Both changes together – a revised start date and the correct NOx data – result in an increase in NOx emissions associated with the LEP of **409 tons** (2103-1694=409). This increase exceeds the NOx significance threshold by a significant amount and is within my estimate range of NOx increases and those of CEMEX’s consultant, MMA.

Third, Mr. Hofmann’s calculation only considers NOx emission increases from the kiln. The LEP included increases in quarry output, which has to be dried. The Lyons facility includes a gas-fired dryer. The NOx emissions from this dryer would have increased, contributing to the LEP NOx increase. These emissions are important to include, as Mr. Hofmann’s estimate of the actual NOx increase is only 5.6 tons shy of the NOx significance threshold.

¹⁶² Hofmann Expert Report, Table 5-7, p. 5-27.

¹⁶³ Hofmann Expert Report, p. 5-27.

¹⁶⁴ Fox Expert Report, p. 5, Table 1; Deposition of Kevin Kelley, December 16, 2010, TR at 154:2 – 5 and 154:20 – 22.

¹⁶⁵ Westlin Rebuttal Report, p. 16.

In sum, even under Mr. Hofmann's limited construction of the LEP, which excludes major parts of the expansion, the LEP increased actual NOx emissions by at least **409 tons**, or ten times the NOx significance threshold.

VI. RAW MILL DEBOTTLENECKING

The raw feed preparation circuit converts limestone, shale, and other ingredients into "raw feed" for the kiln. It includes the raw mill, feed conveying equipment, and silos. This circuit was upgraded and debottlenecked as part of the push to expand plant capacity. The upgrades included a new raw feed transfer compressor, installed in December 1997 to relieve a raw mill limitation,¹⁶⁶ and enlargement of a transfer line.¹⁶⁷

Mr. Hofmann argues that the raw mill changes did not debottleneck the kiln system and that the kiln itself always remained the bottleneck. Thus, increased production at the raw mill would not have resulted in any NOx increases.¹⁶⁸ This is incorrect.

In 1997, CEMEX removed major production limitations at the kiln. The calciner combustion chamber was modified, the horsepower of the main kiln fan motor was increased, and oxygen enrichment via purchased oxygen injected into the tertiary air duct was in routine use.¹⁶⁹ Once the kiln was debottlenecked,¹⁷⁰ CEMEX removed production limitations at

¹⁶⁶ 1998 Capital Budget, US00000019.

¹⁶⁷ Lyons Clinker Production History, May 17, 2000, CMXLYO00045592; Email from John Lohr to Dave Repasz, et. al, September 19, 2000, Re: Power Review.xls, CMXLYO00045547; Lyons Plant Expansion – Phase I, Project Scope – Revised 12/15/97, CEMEX_NEIC_00003299 at 3301.

¹⁶⁸ Hofmann Expert Report, p. 1-6, Opinion 19 and Sec. 5.4.2, p. 5-24.

¹⁶⁹ Deposition of Andrew Jackura, September 16, 2010, TR at 76:1 – 3.

¹⁷⁰ A bottleneck is a limitation on the operation of an emission unit due to restrictions at upstream or downstream units that prevents it from reaching its full capacity. *See also*, Deposition of John Lohr, October 11, 2010, TR at 140:25 – 141:6 ("Bottleneck, to me, is something that will limit or has the potential to limit our production, and I believe that that particular term is used in EPA regulations..."), 191:6 – 11 ("...It means eliminating the limitations.").

Deposition of Keith Baugues, July 30, 2010, TR at 13:13 – 25:

I guess in the term of air pollution, the throughput going to this particular unit as hampered by the kiln. In other words, it could have done – the entire overall process could have made more material if the kiln could have operated at a higher capacity or better throughput. Something in the kiln was not operating correctly to give them an increased input. Something they must have changed in '97 changed that, therefore, as I say, they could produce more tons of cement per hour.

Deposition of Ram Seetharam, August 11, 2010, TR at 23:9 – 24:11:

See, if a facility has several – several processes, say, and then one is dependent on the previous, that one is dependent on the previous one, and some of these sub processes may have different

upstream and downstream material handling facilities – the quarry, crusher, separator, and finishing mill. The 1998 capital budget, submitted in December 1997, noted: “Plant capacity is currently limited by raw mill and finish mill capacity. If these systems were upgraded, the plant could burn more oxygen (up to 85 tons per day) to produce more clinker.”¹⁷¹ The 1998 capital budget provided for the purchase and installation of equipment for the raw feed preparation system and the finish mill system to support up to 600,000 ton/yr of cement production.¹⁷² The January 1998 Report of Operations explained that raw feed preparation had been “the recent clinker production bottleneck.”¹⁷³ Mr. Lohr, in a January 1998 memorandum, also stated that the kiln had been the bottleneck for most of the plant’s history, but “improvements in kiln operation in 1997 reversed that situation. With the combustion chamber improvement and the injection of liquid oxygen, the kiln began to outrun the upstream [raw mill] and downstream [finish mill] systems.”¹⁷⁴ The September 21, 1998 project scope for the Lyons Expansion Project noted that even after another compressor was installed on the raw mill transfer system in late 1997,¹⁷⁵ “[t]he bottleneck of the raw mill system is still the feed conveying equipment.”¹⁷⁶ It thus proposed to increase the raw mill FK transport line size to reduce the backpressure so the feed conveying equipment would no longer be a restriction on raw mill performance.¹⁷⁷ The 1998 Lyons Plant Performance Summary for 1998 explained:

At the beginning of the year the kiln output was limited by raw mill and crusher/dryer system performance and liquid oxygen availability. The plant has been able to improve raw mill rate and reliability to the point where it is no longer a production bottleneck. This has been done with the installation of an additional

design rates or production capacities and some may have much lower than those – the other processes, so when –when the facility decides that this particular one has become a constraint for their – for their entire facility, if they decide to do a modification of that particular process, make some changes in the equipment, make some maybe – maybe an augmentation of the particular one to sort of increase – increase a particular one, this process is know as debottlenecking, in a sense, that would be subject to modification reviews...the facility itself, its potential to emit have increase because of this particular debottlenecking.

The increase in emissions that result from removing bottlenecks are included in the emission netting process used to determine applicability of New Source Review. See NSR Manual, p. A.37 and A.38.

¹⁷¹ 1998 Capital Budget, US00000019.

¹⁷² *Ibid.*

¹⁷³ Lyons Plant – Monthly Report of Operations, January 1998, CMXLYO00050091 at CMXLYO00050143.

¹⁷⁴ Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, Re: Lyons Kiln Production and Oxygen Use, January 4, 1998, US00000020 at 21.

¹⁷⁵ US00000020 at 21 (“The plant installed another compressor on the raw mill transfer system in late 1997. This change, in combination with the smaller rock going to the raw mill, has improved rates on this system up to 12% demonstrated in January.”).

¹⁷⁶ Lyons Plant Expansion, Project Scope – Revised 9/21/98, CMXLYO00155865 at 155867.

¹⁷⁷ *Ibid.*

transfer compressor and a great deal of focus on system uptime and rate. Improvements were made to the crusher/dryer system as well. Although uptime improved by 5% and throughput improved by 7% for the year, this system remains a system bottleneck. Additional changes to the bleed-in dampers and operational parameters are being implemented in December, 1998 which should further improve system results.¹⁷⁸

The AFE submitted in February 1999 for the 1999 Capital Budget includes an item captioned “Lyons Capacity Increase (Carry Over)” to “[p]urchase and install equipment for the raw feed preparation system and the finish mill system to support up to 600,000 t/yr cement production [the original expansion goal]. High efficiency separator, Raw mill compressor, Raw mill dust collector modifications, Raw mill air heater, Cooler Upgrade.”¹⁷⁹

In October 2000, CEMEX retroactively summarized its actions concerning the raw mill system as follows: “We made conscious changes in raw mill operation beginning early in 1997 with the modification of the calciner combustion chamber. Immediately after this change we purchased a higher horsepower motor for the ID fan. These two changes combined produced about a 10% increase in kiln throughput.”¹⁸⁰ CEMEX further explained:

Our constraints on throughput on the raw mill have historically been related to kiln feed transfer capacity (pneumatic transfer from the raw mill to the kiln feed silo). We have changed some hardware on the system to improve our throughput and keep up with kiln capacity. Over the last 2 years, we have added a compressor and enlarged the transfer line. This has increased raw mill production and transfer rates from about 120 tph to about 140 tph.¹⁸¹

Thus, the record shows that the raw mill debottlenecked the kiln. Changes were made to remove these bottlenecks, allowing the raw mill to make more feed and the kiln to produce more clinker. Thus, emissions from debottlenecking the raw mill system should have increased clinker production and should have been included in the PSD applicability determination.

VII. THE 1997 OXYGEN ENRICHMENT TEST

In early March 1997, CEMEX conducted a test to determine if oxygen enrichment could be used to increase clinker production (the “1997 OE Test”).¹⁸² The results of this test are

¹⁷⁸ Southdown, Inc., Lyons Plant Performance Summary, 1998 Operating Year, CMXLYO00038988 at 3898989.

¹⁷⁹ Lyons Plant, August 28, 1998, CMXLYO00155783 at 155787.

¹⁸⁰ Email from John Lohr to Dave Repasz, et. al, September 19, 2000, Re: Power Review.xls, CMXLYO00045547.

¹⁸¹ *Ibid.*

¹⁸² Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00612385; Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155868.

described in a trade publication¹⁸³ and in a patent.¹⁸⁴ The purpose of this 1997 OE Test was to determine the increase in clinker production that could be achieved by injecting oxygen at various points in the pyroprocessing system. At the time, it was believed that oxygen injection into the kiln increased thermal NOx.¹⁸⁵ Thus, NOx emissions were included as a test variable, but the focus of the test and CEMEX's analysis of it were on clinker production.¹⁸⁶ Both Herman Tseng and John Lohr believed that the test showed that oxygen enrichment did not change NOx emissions ("No significant change in NOx emissions was detected.").¹⁸⁷

Mr. Hofmann and Dr. Davis reanalyzed this data, concluding that NOx emissions on both a pounds per hour and pounds per ton of feed basis were lower when oxygen was injected, compared to when it was not. This occurred even though both kiln feed rate and calciner coal feed rate increased when oxygen was injected, compared to when it was not.¹⁸⁸ This counterintuitive result led Mr. Hofmann to admit: "We are not aware of any documented explanation of precisely how oxygen injected into the calciner reduces the overall NOx generated in the process."¹⁸⁹

In my opinion, CEMEX is unable to explain the conclusion reached by its experts because Mr. Hofmann and Dr. Davis both failed to take into account the confounding effect of kiln temperature, which resulted from manipulation of the kiln system during the 1997 OE Test to optimize the generation of clinker.¹⁹⁰

¹⁸³ Herman Tseng & John W. Lohr, Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00612451.

¹⁸⁴ Herman H. Tseng & Philip A. Alsop, Apparatus for Oxygen Enrichment of Cement Kiln System, U.S. Patent No. US 6,688,883 B2, February 10, 2004, Patent Application, February 10, 2004, CMXLYO00628933.

¹⁸⁵ Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00612385 (Item #4: "Increased (thermal) NOx formation in kiln.").

¹⁸⁶ 1997 Oxygen Test Spreadsheet, CMXLYO00612454; Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00612385 at 612386 ("The principal purposes of the test are: (1) To confirm that 2% of the combustion air can be safely and beneficially replaced by oxygen and, if possible, to test the estimate whether higher levels would be possible. (2) To determine the incremental production increase in terms of Tons clinker per Ton oxygen.") and 612387 – 612405 (various analyses of data).

¹⁸⁷ Herman Tseng & John W. Lohr, Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00612451 at 612152.

¹⁸⁸ Hofmann Expert Report, p. 5-16; Davis Expert Report, p. 14 – 15.

¹⁸⁹ Hofmann Expert Report, p. 5-16 to 5-17.

¹⁹⁰ Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00402688 at 402692 – 402693 ("amped out on ID Fan (1029-1031 RPM)" and "As oxygen is injected, the operators have been instructed to maintain these process conditions and to increase feed and fuel as if we suddenly had additional ID Fan."); see Deposition of John Lohr, TR at 60:4 – 17 and 61:17 – 23 as to "amped out."

It is well known that the formation of NO_x in cement kilns increases exponentially with temperature.¹⁹¹ As temperatures in the calciner are typically about 1500-1600 °F,¹⁹² while temperatures in the kiln are much higher, at least 3,400 °F,¹⁹³ a much larger fraction of the NO_x originates in the kiln. Temperature in the combustion zone of the kiln where the thermal NO_x is formed is difficult to measure accurately. It is often measured with optical pyrometers that provide relative values so that the operator can assess changes in temperature. The absolute value of the combustion zone temperature is not particularly relevant to the operator, and it is seldom recorded and archived. If measured during the 1997 OE Test, it was not reported in the daily Operations Reports and other produced records of this test. Instead, surrogates for burning zone temperature were reported.

Operators use kiln amps as an indicator of kiln burning zone temperature.¹⁹⁴ Kiln amps, or torque, are the amps drawn by the motor that turns the kiln.¹⁹⁵ Thus, kiln amps are an indirect measure of NO_x emissions.¹⁹⁶ Kiln burning zone temperature is the cause and kiln amps are the effect. If amps are high, the operator will increase feed; if amps are low, feed will be decreased. Kiln amps increase when kiln burning zone temperatures increase.¹⁹⁷ Conversely, kiln amps

¹⁹¹ From United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW: Gerald L. Young, Penta Engineering, NO_x Formation in Rotary Kilns Producing Cement Clinker Applicable NO_x Control Techniques and Cost Effectiveness of these Control Techniques, IEEE Cement Industry Technical Conference, 2002, USPROD01-003473 at 3474, Sec. 2.0, and 3482; Deposition of Andrew Jackura, February 27, 2008, TR at 49:16 – 56:22. Zephyr Environmental Corporation, Summary of Control Techniques for Nitrogen Oxides, Portland Cement Association, PCA R&D SN3049, 2008, p. 13. Available at: <http://www.cement.org/bookstore/download.asp?mediatypeid=1&id=16594&itemid=SN3049>.

¹⁹² From United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW: KHD Humboldt Wedag, Operating Instructions for Rotary Kiln with Single-String Cyclone Preheater PR 73(76)52 PYROCLON with PYROTOP Coal Dust Firing, USPRODCONT-0000221 at 239.

¹⁹³ Zephyr Environmental Corporation, Summary of Control Techniques for Nitrogen Oxides, Portland Cement Association, PCA R&D SN3049, 2008, p. 7, 13. Available at: <http://www.cement.org/bookstore/download.asp?mediatypeid=1&id=16594&itemid=SN3049>.

¹⁹⁴ Deposition of Kevin Kelley, TR at 87:12 – 87:14 (A. I'm looking at the drive amps. That would be an indication of the burning zone of the temperature, which shows that it would be a little bit hotter.).

¹⁹⁵ Deposition of Randal Wiley, November 11, 2010, TR at 115:24 – 116:8 (“Q. Okay. Let me ask you about kiln amps. When kiln amps are high -- what are kiln amps? A. It is the amperage that the motor is drawing to turn the kiln, so it is the force required to turn the kiln. Q. So when kiln amps are high, it takes more -- it is because more force is necessary to turn the kiln at a constant rate? A. Correct.”)

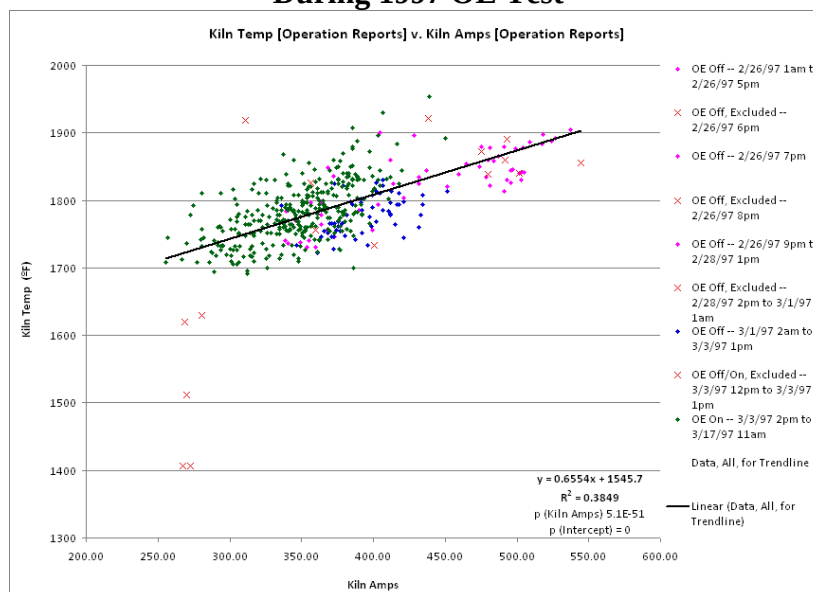
¹⁹⁶ Deposition of Kevin Kelley, December 16, 2010, TR at 49:8 – 21:

A. In reference to the feed, I mean, just the feed is not what varies the NO_x. It's adjusting the speed is going to vary temperatures in the system, which the temperature in the system is what causes a variance in the NO_x. Q. Okay. Kiln speed? A. Again, kiln speed, like I say, usually refers to kiln speed and kiln feed together. But if you change kiln speed independently of the kiln feed, then that's going to change even later and move more material into the burning zone; or you are going to slow the movement of material from one end of the kiln to the discharge end, which is going to affect the temperatures, which has effect on NO_x.

decline when kiln burning zone temperatures decline.¹⁹⁸ Kiln burning zone temperature, in turn, is manipulated by altering the kiln feed and kiln speed.¹⁹⁹

The relationship between kiln temperature (as indicated by the variable, kiln feed hood temperature, at the back end of the kiln) and kiln amps is shown in Figure 10. This figure plots the same subset of the 1997 test data as analyzed by Mr. Hofmann (baseline + Part 2, when oxygen was injected into the tertiary air duct).

Figure 10
Kiln Feed Hood Temperature vs. Kiln Amps
During 1997 OE Test



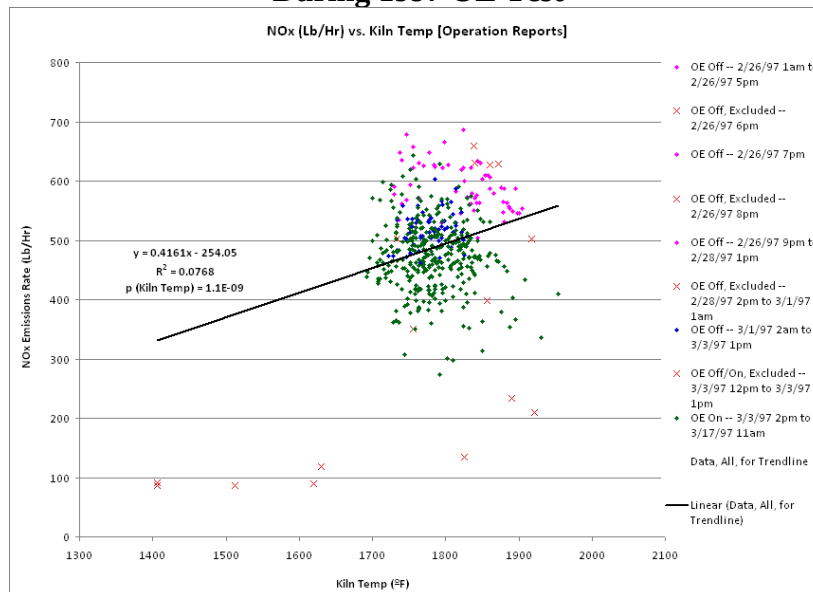
¹⁹⁷ Deposition of John Lohr, October 11, 2010, TR at 243:24 – 244:3 (“If your amps are trending up, the kiln is heating up. If your amps are trending down, your kiln is losing heat...It’s an indicator of kiln stability, a crude indicator of kiln stability.”); Deposition of Kevin Kelley, December 16, 2010, TR at 86:1 – 86:5 (A. Yes. That’s my -- As I said earlier, the idea was to heat the kiln up; and so you’re going to be operating at higher temperatures and higher kiln amps and to have a stable kiln during oxygen on and off.); 87:2 – 14 (“These -- The temperatures here, it depends on what the burning zone temperature is. So these are referring to the combustion chamber temperature and the feed hood temperature and is not indicating the burning zone temperature...I’m looking at the drive amps. That would be an indication of the burning zone of the temperature, which shows that it would be a little bit hotter.”).

¹⁹⁸ Lyons Plant Pyroprocess Training, CMXLYO00186916 at 186989 (“Bottom Line: The hotter the burning zone, the higher the NOx”), 187037 (kiln amps one of levers to control burning zone), 187038 (“The colder the [burning zone], the lower the amps”); Deposition of Randal Wiley, November 10, 2010, TR at 115:1 – 115:3; Kurt E. Peray, *The Rotary Cement Kiln*, Classical Publishing Co., 1986, p. 239 – 40 (NOx is one of the proven indicators of changes in burning-zone temperatures), 246 (NOx increases when the burning zone is heating up and decreases when it is cooling down). See also, Greer Rebuttal Report, June 24, 2011.

¹⁹⁹ Deposition of Kevin Kelley, December 16, 2010, TR at 49:8 – 21.

Increases in kiln burning zone temperature increase NO_x emissions, and conversely, decreases in kiln burning temperature decrease NO_x emissions, as illustrated by Figure 11. This figure plots kiln feed hood temperature (at the back end of the kiln), an indicator of kiln burning zone temperature, against NO_x emissions for the same subset of the 1997 test data as analyzed by Mr. Hofmann (baseline + Part 2). Kiln burning zone temperature itself was not measured.

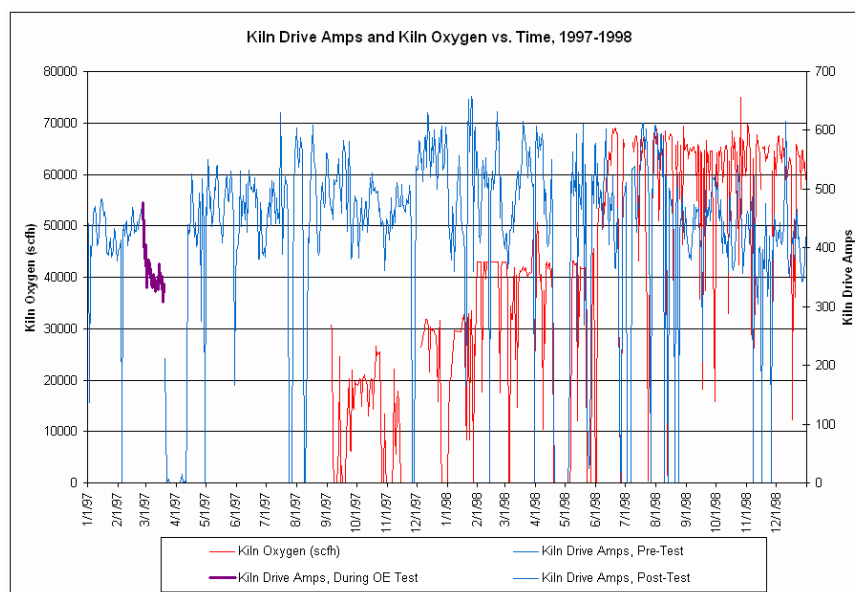
Figure 11
NO_x Emissions (lb/hr) vs. Kiln Feed Hood Temperature
During 1997 OE Test



Kiln burning zone temperature, in turn, normally is manipulated using kiln speed and the flame at the discharge end of the kiln. Other operating parameters, such as the concentration of oxygen at the feed end of the kiln (which may indicate the quantity of secondary air drawn past the flame), can be manipulated to affect the burning zone temperature.

During the oxygen injection test, feed and fuel were increased to determine how much additional clinker could be produced. This reduced kiln burning zone temperature, indicated by the decline in kiln amps, which decreased NO_x emissions. While kiln burning zone temperature was not reported in the daily Operations Reports, kiln amps were recorded. Kiln amps dropped during the 1997 OE Test from an average of **412** (268-545) during the oxygen off portion of the test (the baseline) to **343** (254-450) during the oxygen on portion of the test. The kiln amps during the oxygen off portion of the test were lower than normal operating levels both before and after the 1997 oxygen test. This is shown by Figure 12, which plots kiln amps overlaid on the amount of oxygen injected from Daily Operation Reports for 1997 to 1998. The oxygen-on test period is shown in purple to distinguish it from normal operating conditions. Increasing feed and fuel while decreasing kiln burning zone temperature causes NO_x emissions to decline, regardless of the amount of oxygen injected.

Figure 12



Thus, the 1997 OE Test was not representative of normal operating conditions and cannot be used to determine the impact of oxygen enrichment on NO_x. Normal operating conditions were manipulated during the test to maximize the amount of clinker produced. This caused kiln burning zone temperature as reflected by kiln amps and NO_x emissions to decline. Kiln burning zone temperature (as indicated by kiln amps) did not decline in the future when oxygen was injected. The manipulation of fuel, feed, and temperature during the 1997 OE Test to optimize clinker production renders the 1997 OE Test unsuitable for drawing inferences about the effect of oxygen injection on NO_x emissions during normal operation.

Mr. Hofmann realized that there was no documented explanation for how oxygen injection into the calciner could reduce NO_x generation, which led him to speculate as to what possible mechanism(s) might produce this result.²⁰⁰ Data collected during normal operating conditions indicate Hofmann's speculations are wrong.

A complete set of hourly data (NO_x, clinker, oxygen) for 1998 compiled from the records that CEMEX produced indicates the opposite relationship between NO_x emissions and oxygen injection. See Exhibit 1. This much larger data set is not subject to the confounding influence of reduced kiln amps experienced during the 1997 OE Test. Thus, it provides greater confidence in conclusions as to the effect of oxygen injection on NO_x emissions.

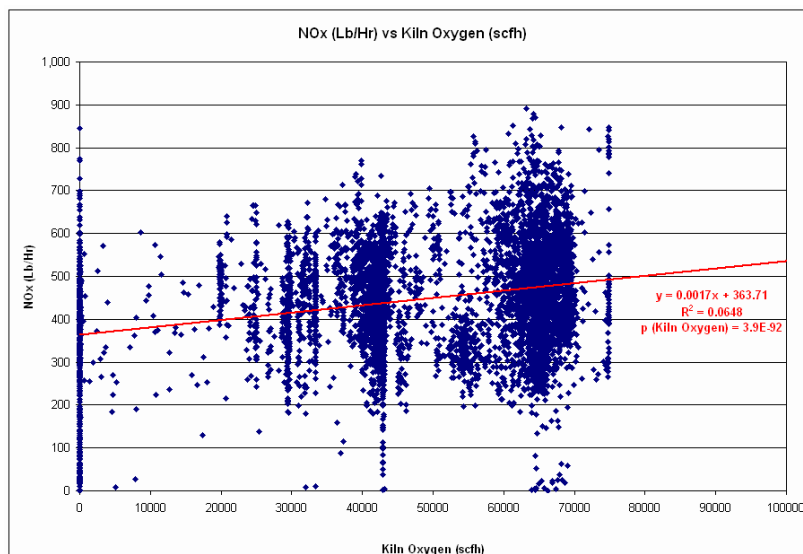
The 1998 NO_x CEMS data (lb/hr) is plotted in Figure 13 against the amount of oxygen injected (scfh).²⁰¹ I used linear regression to fit a straight line to this data. My analysis shows

²⁰⁰ Hofmann Expert Report, p. 5-16 to 5-17; Davis Expert Report, Sec. 3.1, p. 12.

²⁰¹ This figure excludes all data "flagged" in CEMEX's original files. This figure differs slightly from a similar figure in the Rebuttal Expert Report of Bradley Venner, Ph.D., United States v. CEMEX, Inc., Civil Action No. 09-

that NO_x emissions in lb/hr increase in direct proportion to the amount of oxygen that is added. Although oxygen injection explains only 7% of the variability ($r^2 = 0.07$) in NO_x, the slope of the line is positive (0.0017) and is statistically significant ($p = 3.9E-92$).

Figure 13



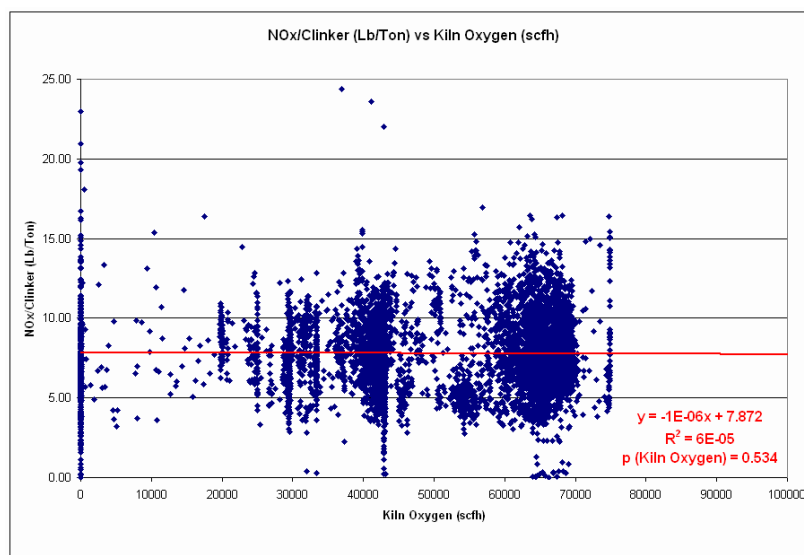
I also analyzed the hourly 1998 CEMS data by calculating the NO_x emission factor in pounds of NO_x per tons of clinker. See Exhibit 1. The results of this analysis are summarized in Figure 14. This figure indicates the amount of oxygen injected does not significantly impact the NO_x emission factor in lb/ton. The correlation coefficient (r^2) for this linear regression is 0.00006, a vanishingly small value which indicates oxygen injection explains almost none of the variability in NO_x emissions expressed in lb/ton. The fitted line is nearly parallel to the x axis, trending slightly downward. The slope is not statistically significant ($p = 0.5$) and amounts to only 0.01 lb NO_x per ton clinker for each 10,000 scfh of oxygen injected. Injection of 82,000 scfh of oxygen, the maximum projected use,²⁰² would reduce the NO_x emission factor that I used in my netting analyses by only 0.08 lb/ton. This would reduce the baseline emission factor I used for scenarios 1 and 3²⁰³ from 8.1 lb/ton to 8.0 lb/ton and for scenarios 2 and 4 from 7.6 to 7.5 lb/ton. This does not change the results of my netting analysis.

cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011 (hereinafter "Venner Rebuttal Report"), due to the use of different exclusion criteria. The results are the same.

²⁰² Lyons Plant – Monthly Report of Operations, August 1999, CMXLYO00042970 at 42971 ("The new VSA oxygen plant began operation in late August. It is currently generating the equivalent of about 52 tons per day of pure liquid oxygen. In addition to the VSA production, Lyons ran about 30 tons/day of liquid oxygen which will continue through September."); Daily Production Notes for 2/26/99 – 12/17/99, CMXLYO00095456 – CMXLYO00095725, indicate continuing routine oxygen injection and the use of both VSA plant oxygen and imported oxygen after the new plant came online in mid-August (8/14/99).

²⁰³ Fox Expert Report, p. 14, Table 2.

Figure 14



In sum, the 1997 OE Test cannot be used to determine the impact of oxygen injection on NOx emissions due to the confounding influence of reduced kiln burning zone temperature (as indicated by kiln amps). An entire year of hourly data indicates that NOx emissions in lb/hr increase when oxygen is injected. A slight statistically insignificant decline occurs in the NOx emission factor in lb/ton.

VIII. PERMITTING

The Lyons plant had historically operated under four permits: quarry, crusher, dryer, and kiln.²⁰⁴ In 1998, the particulate matter emissions from material handling changes required to support the clinker production increase were permitted.²⁰⁵ This resulted in revisions to the quarry and crusher permits and in new permits for the finish mill and CKD handling system.²⁰⁶ However, CEMEX did not at that time apply for permits for those parts of the 1997 to 2000

²⁰⁴ Several of the remarks made by Mr. Hofmann throughout his report which address the permitting process and CEMEX's interactions with the CDPHE may not be appropriate for inclusion in his expert report. As that matter will be decided sometime after the June 24, 2011 deadline for the United States submission of its rebuttal reports, counsel for the United States has asked that, out of an abundance of caution, I rebut those portions of Mr. Hofmann that I find objectionable. As such this rebuttal shall not be construed to accede in any way to the proposition that Mr. Hofmann's inclusion of material which address the permitting process and CEMEX's interactions with the CDPHE in his expert report is appropriate or to waive the United States' right to challenge the same.

²⁰⁵ Southdown, Inc., Construction Permit and Permit Modification Application, April 30, 1998, CMXLYO00015334 at 15582 (Overview of all permits); Colorado Department of Public Health and Environment – Air Pollution Control Division (CDPHE – APCD), Construction Permits: Permit No. 98-BO-0259, September 11, 1998, CEMEX_NEIC_00000246 (New finish mill); Permit No. 98-BO-0292, September 11, 1998, CEMEX_NEIC_00000229 (New raw materials storage and handling at the plant site); Permit No. 93-BO-1414-F, September 11, 1998, CEMEX_NEIC_00000257 (Modifications to Dowe Flats Quarry).

²⁰⁶ Permitting history discussed in Sec. 3.1 of Hofmann Expert Report, p. 3-1 to 3-2.

expansion project that debottlenecked the kiln. Mr. Hofmann argues that permitting changes to the pyroprocessing system was unnecessary because: oxygen injection did not contribute to NOx emissions increases;²⁰⁷ it was reasonable for CEMEX to rely on CDPHE's alleged determination concerning oxygen enrichment;²⁰⁸ CEMEX did not mislead CDPHE;²⁰⁹ and MMA advised CEMEX that it need not obtain permits.²¹⁰ I address each of these points below to show how in my opinion, Mr. Hofmann has seriously misinterpreted the record in this matter.

A. Oxygen Enrichment Contributed To Increased Clinker Production Which Caused Additional NOx Emissions.

Mr. Hofmann asserts that PSD did not apply because oxygen enrichment was “the key to achieving planned clinker production increases” and “[e]xperiments demonstrated that OE injection did not contribute to any NOx emission increases but had capability of achieving clinker production increases.”²¹¹ He further claims oxygen injection actually causes NOx to decline²¹² and that “PSD does not apply unless the source finds that there is a causal link between the proposed change and any post-change increase in emissions.”²¹³

Mr. Hofmann's contention that oxygen injection causes NOx in both lb/hr and lb/ton to decline, upon which he bases his assertion that no causal link exists between oxygen injection and post-change increases in NOx, is based on his interpretation of the 1997 OE Test.²¹⁴ That interpretation is incorrect, as explained in Section VII, and is inconsistent with his own theory of variability. As explained elsewhere in Section VII of this rebuttal and in the Expert Rebuttal Reports of Bradley Venner, Ph.D., and Walter Greer, the 1997 OE Test cannot be used to draw any inferences about the causal relationship between oxygen injection and NOx emissions for the following reasons: the 1997 OE test conditions do not represent conditions experienced during normal kiln operations; correctly interpreted, the 1997 OE test results do not support a

²⁰⁷ Hofmann Expert Report, p. 5-25 to 5-26.

²⁰⁸ Hofmann Expert Report, Summary of Opinions, p. 1-6, Opinion No. 17.

²⁰⁹ Hoffman Expert Report, p. 5-23 (“CEMEX management presented detailed project information to the CDPHE in the form of a PowerPoint presentation on March 6, 1998. Nowhere in this presentation is there any evidence that CEMEX management made “statements... to state or local permitting agencies about the source's plans for operation...show[ing] intent to circumvent preconstruction review requirements...”)

²¹⁰ Hofmann Expert Report, Summary of Opinions, p. 1-6, Opinion No. 18.

²¹¹ Hoffman Expert Report, p. 5-25.

²¹² Hofmann Expert Report, p. 5-25, based on Sec. 5.3.1, beginning at p. 5-11.

²¹³ Hofmann Expert Report, p. 5-26.

²¹⁴ Hofmann Expert Report, p. 5-25 to 5-26 and Sec. 5.3.1, beginning at p. 5-11.

conclusion that NOx emissions are reduced by oxygen injection; the 1997 and 1998²¹⁵ OE Tests contradict each other; and the 1997 OE test itself, by virtue of its design, is insufficient to assess whether a causal relationship exists between oxygen injection and NOx emissions.

Further, as explained in Section VII and illustrated in Figure 13, a full year of CEMEX operating data for 1998 indicates that oxygen injection likely increased NOx emissions measured in lbs/hr.

Finally, if we hold CEMEX to the representation it made to CDPHE in April 1998, “NOx emissions are not influenced by the presence of OE,”²¹⁶ then, applying a constant emission factor, the increase in production associated with oxygen enrichment alone would have triggered regulatory thresholds under PSD.²¹⁷

B. Reliance on the Determination of a Regulator Is Unwarranted When Incomplete Information Has Been Provided to that Regulator

Mr. Hofmann asserts:

The air permits issued by CDPHE to the Lyons plant allowed production increases associated with the LEP (other upstream and downstream changes). CDPHE agreed with CEMEX’s position that the kiln NOx emissions would not increase as a result of OE. It was reasonable for CEMEX to rely upon CDPHE’s position and not to undergo PSD review. Accordingly, in September 1998, CDPHE issued 2 revised permits (crusher and quarry) and 3 new permits (raw material handling, finish mill system, and CKD handling).²¹⁸

As discussed below, it was not reasonable for CEMEX to rely upon CDPHE’s “position.”

Oxygen injection was only one of the tools in the expansion tool chest. Other changes contributed significantly to the increased production goal. CEMEX’s own tabulation indicates the new oxygen plant contributed only 16% of the targeted clinker production increase. Thus, even assuming oxygen injection had no impact on NOx, other parts of the expansion that CEMEX did not disclose to CDPHE did cause NOx to increase. Moreover, it is clear that during

²¹⁵ Deposition of John Lohr, October 11, 2010, TR at 242:21 – 22 (Mr. Lohr discussing Dr. Boardman’s assessment of the 1998 OE Test concerning the effects of oxygen enrichment on NOx emissions: “A. His conclusion was there is not statistically significant difference” on NOx emissions between oxygen on and oxygen off.).

²¹⁶ Letter from John Lohr, Plant Manager, CEMEX, to Ram Seetharam, CDPHE –APCD, April 29, 1998, CEMEX_NEIC_00008478 at 8479.

²¹⁷ Mr. Hofmann seems to ignore the fact that oxygen enrichment was only one component of the project CEMEX implemented at the Lyons facility between 1997 and 2000. Other changes contributed significantly to the increased production goal. CEMEX’s own tabulation indicates the new oxygen plant contributed only 16% of the targeted clinker production increase. Thus, even assuming oxygen injection had no impact on NOx emissions, other parts of the expansion project did cause NOx to increase above regulatory significant thresholds.

²¹⁸ Hofmann Expert Report, p. 1-6, Opinion 17.

the 1998 permitting process, CDPHE's understanding of CEMEX's use of oxygen was incomplete, due to CEMEX's omissions. In 1998, during the permitting process, CEMEX never told CDPHE that in 1997, it had shifted from experimentation to routine use of oxygen injection. Mr. Cooley, assistant to CEMEX's Corporate Director of Air Permitting, testified:

Q. [A]t any of the meetings that you attended where CDPHE was present, did CEMEX say to CDPHE, "We're no longer experimenting with oxygen. It's part of our routine operation of the plant's pyroprocessing system"? A. I don't recall that ever being said at a meeting.²¹⁹

CDPHE's permitting engineer clearly did not understand that at the time CEMEX presented its 1998 OE test results, April 29, 1998, CEMEX was routinely injecting oxygen.

Q. If Cemex were to believe that the only way they could have increased clinker production from the kiln was through the injection of oxygen, didn't they, in fact, follow through with coming back later when they presented their -- their data that oxygen injection did not increase emissions? A. See, the -- see, the information that we have, of course, at this time is that they did implement the injection project. At the time that we had those discussions, subsequent discussions, later when processing this permit, we still did not have any information that they were implementing those projects.²²⁰

CDPHE believed routine use would not commence until after the VSA plant was constructed.

Q. Did you suspect that Cemex might not be routinely injecting oxygen until after this plant was built? A. That was still in the sort of proposal stage, regular, so at the time we had the meeting, it was only the -- it was only the experimenting. Q. That was your understanding based on your discussions? A. That's right, they were experimenting with the source.²²¹

Further, CEMEX never told CDPHE that oxygen injection would result in increased clinker production. This is evident in deposition testimony of Mr. Seetharam (State permit engineer), Mr. Cooley, and Mr. Goodrich (present Plant Manager):

"Q. So they never told you it [oxygen injection] was to increase production; is that right? A. That's right."²²²

²¹⁹ Deposition of Bradley Cooley, September 20, 2010, TR at 57:10-15.

²²⁰ Deposition of Ram Seetharam, August 11, 2010, TR at 227:12-25.

²²¹ Deposition of Ram Seetharam, August 11, 2010, TR at 99:11-21.

²²² Deposition of Ram Seetharam, August 11, 2010, TR at RS 101:17-19.

Q. In any of the meetings that you attended with the State, did you ever hear CEMEX explain to the State whether and to what degree oxygen enrichment would increase clinker production? A. No. Q. No, you never heard them have that discussion with the State? A. No, I don't recall them ever having that specific discussion with the State.²²³

Q. Do you know whether in the context of the March '97 test or the March '98 test any information was provided to Boulder County Health, CDPHE or EPA that would demonstrate the impact oxygen enrichment might have on clinker production? . . . A. No.²²⁴

Finally, CDPHE did not view the April 1998 OE test information to be part of any permit determination.

Q. So their ultimate conclusion is that NOx emissions are not influenced by the presence of oxygen injection? . . . Q. Did the state accept this conclusion? A. See, as I said, these are not considered as applications for a modification. Unless an application comes with this, we will not do any analysis and send a response, so they are just some information that they send. There are so many companies, they just send information or so many of their -- see, we don't do any of those analysis unless it is submitted as a -- as an application and an application notice. Q. Well, Cemex got the idea that based on this letter and this testing, that it was okay with the state to go forward with the oxygen injection process because their testing had showed that NOx emissions aren't influenced by oxygen injection. Is that -- is that a mischaracterization of or a misunderstanding of what the state told them? A. See, it is their position that there is no increase, but obviously we have not done any analysis on those, and with this information no analysis can be done, too, because there is no other information except these numbers here. It does not make any sense. . . .²²⁵

Having misrepresented both the timing and purpose of oxygen injection, CEMEX had no reason to rely on CDPHE's actions. Further, given the problems with the 1998 OE test, the information CEMEX presented to CDPHE related to that test was insufficient for any kind of informed determination.²²⁶

The New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting, Draft October 1990, page A.36, provides:

²²³ Deposition of Bradley Cooley, September 20, 2010, TR at 78: 1 – 9.

²²⁴ Deposition of Stephen Goodrich, December 9, 2010, TR at 102:3-13.

²²⁵ Deposition of Ram Seetharam, August 11, 2010, TR at 148:2 – 150:24 (referencing Seetharam Exhibit 72, April 28, 1998 "Boardman" Letter) (emphasis added).

²²⁶ See Greer Rebuttal Report.

A deliberate decision to split an otherwise “significant” project into two or more smaller projects to avoid PSD review would be viewed as circumvention and would subject the entire project to enforcement action if construction on any of the small projects commences without a valid PSD permit.

As explained below, this is what CEMEX did when it approached the CDPHE in January 1998 to initiate permitting for only the raw materials handling portion of the expansion project, while omitting those changes that it made, and planned to make, to the kiln.

Early on, CEMEX was advised by MMA that the quarry, separator, and kiln projects should be considered as one permitting action.

A. Southdown was considering whether they could permit the changes that they were asking for at the quarry themselves and look at any other changes, the separator or the kiln or the dryers or any other things as a separate permitting action, and my conclusion was that those were tied together, and they really couldn't do that. But they were looking whether that was a possibility. Q. So your conclusion was that the plant, both plant and quarry should be permitted together as one? A. They were contemporaneous actions. They were occurring at the same time, so they needed to be permitted as one action. Q. Did you advise CEMEX of your opinions? A. Yes, I did.²²⁷

When asked why permitting only for the raw materials handling portion of the project went forward, Mr. Baugues stated:

Q. And so when you submitted the application, it was only for the upstream and downstream operations; is that correct? A. That's correct. Q. And how did you reach the conclusion that you could submit that operation not including the middle stream, kiln, clinker cooler, and dryer? A. Jit Gill had convinced us that there would be no debottlenecking, therefore no increase in throughput, no increase emissions, so you don't have to this. And that is what the application reflects.²²⁸

Mr. Baugues explained that had MMA known about the debottlenecking activities undertaken by CEMEX in 1997, MMA would have considered them in the permitting process.²²⁹ Thus, the very individual who assisted CEMEX in the preparation of its quarry, crusher, raw mill, and finish mill permits (the upstream and downstream operations) told CEMEX that plant and quarry changes should be permitted together. However, because MMA was not informed of CEMEX's 1997 debottlenecking changes,²³⁰ it did not insist on incorporating plant capacity

²²⁷ Deposition of Keith Baugues, July 30, 2010, TR at 49: 2 – 19 (emphasis added).

²²⁸ *Id.*, TR at 229:15 – 230:15; *see also*, TR at 49: 2 – 19 (emphasis added).

²²⁹ *Id.*, TR at 11:23 – 13:8.

²³⁰ *Id.*, TR at 12:24 – 13:3.

increase changes in the 1998 permitting process. Further, and importantly, this is what Mr. Baugues understood CEMEX to have represented to the CDPHE in its permit application. “As we stated early in the permit, there should have been no [emissions] change from the change in operations, which is the information I was provided by Southdown at the time.”²³¹

As explained in more detail below, the same incomplete picture given to MMA by CEMEX was presented to CDPHE. In short, CEMEX implemented a “deliberate decision to split an otherwise ‘significant’ project into two or more smaller projects to avoid PSD review.”²³²

The produced documents and depositions indicate that the project CEMEX permitted was not the same project CEMEX proposed to build (and in fact had in part built by the time it began seeking permits from the CDPHE in 1998). The project that was permitted²³³ consisted of shutting down two quarries, expanding operations at the Dowe Flats quarry, and replacing existing separators in the finish mill with a new high efficiency separator.²³⁴ The project that was built included many related changes to debottleneck the kiln, including: the 1997 calciner modification; 1997 ID fan motor replacement; commencement of routine oxygen injection in May 1997; CKD return; installation of an annulus boost fan; addition of a new raw mill compressor; and the enlargement of the raw feed transfer line. The impact of these changes on the throughput of the kiln was omitted from CEMEX’s dialogue with CDPHE. CDPHE did not even know that CEMEX was injecting oxygen to increase clinker production.²³⁵ CEMEX only gave CDPHE an incomplete picture of the actions it was taking. CDPHE’s assessment of CEMEX’s oxygen enrichment plans was predicated on that incomplete information.

Moreover, CEMEX used two sets of facts – one for permitting and one to implement its internal expansion plans. This can be determined by comparing the final AFE for the “Lyons Capacity Increase”;²³⁶ other internal CEMEX documents; CDPHE’s notes of meetings;²³⁷ and

²³¹ *Ibid.*

²³² NSR Manual, p. A.36.

²³³ See n. 204 (4/30/98 Application and resulting permits), *supra*.

²³⁴ Deposition of Bradley Cooley, September 20, 2010, TR at 41:2 – 4, 45:23 – 46:1.

²³⁵ See Deposition of Bradley Cooley, September 20, 2010, TR at 78: 1 – 9. (“Q. In any of the meetings that you attended with the State, did you ever hear CEMEX explain to the State whether and to what degree oxygen enrichment would increase clinker production? A. No. Q. No, you never heard them have that discussion with the State? A. No, I don’t recall them ever having that specific discussion with the State.”); Deposition of Ram Seetharam, August 11, 2010, TR at 101:17-19. (“Q. So they never told you it [oxygen injection] was to increase production; is that right? A. That’s right.”); Deposition of Roland Hea, November 2, 2010, TR at 254:8 – 15 (“Q. In any of the material that you reviewed for -- it includes conversations with anyone. Did you find anything that would suggest that CEMEX informed the State, CPDHE, that for every ton of oxygen injected into the pyroprocessing system CEMEX anticipated an increased klinker [sic] yield between 2-1/2 and 3 tons? A. No, I didn't see that.”).

²³⁶ Memorandum from Norris Widener to Karen Krennek, November 18, 1998, Re: Approved AFE, US00000454.

deposition testimony with the April 30, 1998 Permit Application (“4/30/98 Application”).²³⁸ For example, two sets of mix and mining facts were used, one for CEMEX and one for CDPHE.²³⁹ The mix and mining facts provided to CDPHE in the April 30, 1998 Permit Application,²⁴⁰ labeled “Mix and Mining Facts for Submission to the State of Colorado,” assume 450,000 tons of clinker production, thereby suggesting no capacity increase. The mix and mining facts CEMEX relied on, labeled “Anticipated Mix and Dowe Flats Performance After Improvements,” assume 532,224 ton/yr of clinker production, a substantial increase,²⁴¹ consistent with the increase assumed in Mr. Baugues’ netting calculations.

Further, CEMEX physically debottlenecked the “throughput” of the kiln and supporting operations in a series of projects implemented in 1997 – 1998 that were not disclosed to CDPHE. CEMEX modified permit “throughput” limits to allow it to take advantage of the debottlenecking, while asserting kiln “capacity” had not increased. This was achieved by treating the quarry and material preparation, kiln, and finish milling as separate operations with separate permits, while in fact, these operations are interrelated, depend upon one another, and are part of the same stationary source.²⁴²

Additionally, during permitting in 1998, CEMEX did not disclose to CDPHE the clinker production increase or any of the projects that had already been completed and were an integral part of the expansion. Throughout the period that CEMEX worked with CDPHE in 1998, it failed to disclose that in 1997, it had already completed four key projects that were essential to debottleneck the kiln and thus to implement the material handling projects being permitted: (1) combustion chamber modification;²⁴³ (2) larger ID fan motor;²⁴⁴ (3) routine oxygen injection

²³⁷ Handwritten Notes from CDPHE, 1998, US00006339 – 6345 and CMXLYO00015627 at 15629 – 15638; PowerPoint, Southdown, Inc., Meeting with Colorado Department of Health and Environment, APCD, March 6, 1998, US00000594.

²³⁸ Construction Permit and Permit Modification Application, April 30, 1998, CMXLYO00015066.

²³⁹ Memorandum from Gregg St. Clair to Dave Repasz, March 26, 1998, Re: Plant Expansion Project Update, CMXLYO00155748 at 155750, “Lyons Kiln History and Forecast” (Repsz Exhibit 132).

²⁴⁰ Attachment to April 30, 1998, Application, CMXLYO00144605 at 144623, “Attachment A, Future Estimates for Production Rock and Total Material Handled at the Lyons Plant.”

²⁴¹ Memorandum from Gregg St. Clair to Dave Repasz, March 26, 1998, Re: Plant Expansion Project Update, CMXLYO00155748 at 155750, “Lyons Kiln History and Forecast” (Repsz Exhibit 132).

²⁴² Deposition of William Monnett, November 30, 1020, TR at 15:8 – 10 (“Our view of the applicable regulations was that the plant and the quarries were the same major existing source.”), 19:22 – 20:1, 20:13 – 16 (“And our questions was: How are you going to increase it out of the Dowe Flats quarry if you don’t at the same time increase the throughput through the plant?”).

²⁴³ Deposition of Bradley Cooley, September 20, 2010, TR at 49:10 – 14 (“Q. Let me ask you this, returning to the calciner. In any of the meetings with the State CDPHE, did you hear any representative of CEMEX discuss changes to the calciner in 1997 with CDPHE? A. No.”); Deposition of Dennis Meyers, July 21, 2010, TR at 18:4 – 7 (“Q. Were you aware of any change that Cemex may have made to the calciner prior to this [January 6, 1998] meeting? A. No.”).

(which had started in May 1997);²⁴⁵ and (4) larger compressor and piping on raw mill.²⁴⁶ Some of these were mentioned in the initial January 6, 1998 pre-Application meeting as “future projects,”²⁴⁷ when they had already been built. In fact, CEMEX told CDPHE that it had not made any changes in the operation of the kiln since 1993 and identified three of these already completed projects as future projects.²⁴⁸

²⁴⁴ Deposition of Bradley Cooley, September 20, 2010, TR at 77: 18 – 21 (“Q. Did you ever in any of the meetings that you attended hear CEMEX discuss with the State the changes to the induced draft fan motor? A. No.”); Deposition of Dennis Myers, July 21, 2010, TR at 93:20 – 94:6 (“Q. Okay, do you see the next entry, fan horsepower increase? A. Yes. Q. Do you best of your recollection, were you ever informed by Cemex of this change during the time you were discussing the separator and quarry permit with Cemex? A. I don’t recall being informed by Cemex about that. Q. Okay, and by that, you mean you believe Cemex failed to inform the division? A. Yes.”).

²⁴⁵ Deposition of Bradley Cooley, September 20, 2010, TR 57: 10 – 15. (“Q. ... [A]t any of the meetings that you attended where CDPHE was present, did CEMEX say to CDPHE, ‘We’re no longer experimenting with oxygen. It’s part of our routine operation of the plant’s pyroprocessing system’? A. I don’t recall that ever being said at a meeting.”); Deposition of Dennis Myers, July 21, 2010, TR at 94:12 – 95:15 (“Q. Okay. Does it appear from this document that Cemex had employed oxygen enrichment before entering its discussions with the state on the permit, in other words, before the January ‘98 to May ‘98 time period? A. It would appear so, yes. . . . Q. If routinely Cemex was using oxygen enrichment in 1997, would you have expected Cemex to have informed you prior -- prior to the time they did? A. Again, I think that would be reasonable, yes. Q. If, in fact, it was a change in the method of operation, would you have expected them to inform you before making this a routine part of their operations? A. Yes.”); Deposition of Ram Seetharam, August 11, 2010, TR at 106:6 – 13. (“A. No, at the time we don’t even know that they were doing it on a regular basis, so it is still an experimental thing, and they were just evaluating their process. Q. And by at this time, you mean March? A. At the time that we had that meeting. Q. March ‘98? A. Uh-hum.”).

²⁴⁶ Deposition of John Lohr, October 11, 2010, TR at 170:10 – 14. (“Q. Okay. The compressors for the raw mill circuit and the compressor motor, those were not identified to the State in this meeting? A. No, nor did we feel there was any need to do that.”).

²⁴⁷ Deposition of Ram Seetharam, August 11, 2010, TR at 54:23 – 56:7 (“Q. Now, were these changes that they had already done or changes that they planned to do in the future? A. No, these are some of the problems that they just mentioned that they need to look at to achieve their -- to achieve their targets. Q. And your understanding is that they were projects that they were planning to do or thinking about doing in the future? . . . A. Yes. . . . Q. All I want to do is make sure that I understand you correctly, that Cemex presented this item as something that it would -- that it needed to do? A. Yes. Q. And not had already done? A. Yes, sir.”); Seetharam Deposition Exhibit 61, Handwritten Notes from CDPHE, US00006339 at 6344.

²⁴⁸ Deposition of Ram Seetharam, August 12, 2010, TR at 77:12 – 25:

Q. Okay. Let me hand you Exhibit 69 very quickly. Do you see right here in Exhibit 69 where you said Cemex told you there was no change in the operation of the kiln? A. That’s right, that’s during one of their -- the meetings, see, they gave like a history of the -- of the plant, and they said that the modification was done in ‘79, ‘80, and that was what, the calciner and then shorten the kiln. Q. So what Cemex told you in the March 20th, 1998 meeting was that since 1993, there had been no change in the kiln?... A. That’s right.

See also, Seetharam Exhibit 69, Handwritten Notes from CDPHE, US00006339 at 6341.

Mr. Cooley, Jeet Gill's assistant, testified that CEMEX did not tell CDPHE about the 1997 projects.²⁴⁹ When confronted with Mr. Lohr's January 4, 1998 memo on kiln production, Mr. Cooley agreed the 1997 modifications debottlenecked the kiln.²⁵⁰ This starkly contrasts with the rationale Mr. Gill provided to MMA, "Jit Gill had convinced us that there would be no debottlenecking, therefore no increase in throughput, no increase emissions, so you don't have to this. And that is what the [permit] application reflects."²⁵¹ When asked if the information presented by CEMEX to CDPHE in the March 6, 1998 and March 20, 1998 pre-Application meetings was accurate, Mr. Baugues testified in retrospect that it was not. "At the time I couldn't tell because I didn't know about the debottlenecking issue. At this point [after seeing the kiln debottlenecking evidence] I would say it was not."²⁵²

The record above reflects what can reasonably be interpreted as a deliberate decision to split an otherwise significant project into two or more smaller projects (one being the changes to the raw materials handling system and the other being the changes to the kiln and changes that debottlenecked the kiln) to avoid PSD review.²⁵³ CDPHE was never informed that the changes to the kiln were either completed or in progress. As such, CEMEX has no basis for its contention that it was entitled to forego PSD review for the entire project in reliance on CDPHE's decision to permit the changes to the raw materials handling process.

Mr. Hofmann asserts that "[a]n informal memorandum composed by Mr. Dennis Myers of the CDPHE during the same time-frame makes clear the CDPHE was alert to the possibility of potential project aggregation depending on how CEMEX proceeded with subsequent permitting."²⁵⁴ This memorandum was written by Mr. Myers after the initial January 6, 1998 meeting.²⁵⁵ Mr. Myers, the manager of the construction permitting division who was present in the meeting, summarized his impressions in this email. According to Mr. Myers, CEMEX's "bottom line is they need to make more cement." He further notes that CEMEX proposed to initially separate out:

²⁴⁹ Deposition of Bradley Cooley, September 20, 2010, TR at 77: 22 – 25 ("Q. Did you ever hear CEMEX discuss with the State the effect the bottlenecking of the kiln in 1997 had on clinker production? A. No."); Deposition of Roland Hea, November 2, 2010, TR at 77:19 – 78:11 ("Q. Was it the Department's understanding that these future increases in klinker [sic] production would come from modifications CEMEX planned to make to the facility in the future? A. Yes. Q. Did the Department have any understanding that actions taken by CEMEX in 1997 in fact, based on estimates from Mr. Lohr, resulted in an increase in between 10 and 12 percent of production?... A. No.").

²⁵⁰ Deposition of Bradley Cooley, September 20, 2010, TR at 58:19 – 25.

²⁵¹ Deposition of Keith Baugues, July 30, 2010, TR at 229:24 – 230:4; *see also*, TR at 228:23 – 230:15, 49: 2 – 19.

²⁵² Deposition of Keith Baugues, July 30, 2010, TR at 75:1 – 8; *see also*, TR at 83:12 – 19.

²⁵³ NSR Manual, p. A.36.

²⁵⁴ Hofmann Expert Report, p. 5-23 (emphasis omitted).

²⁵⁵ Deposition of Dennis Myers, July 21, 2010, TR at 162:25 – 163:2.

...the permitting of the new separator from increasing production from the kiln, so that they can get the permit for the separator processed more quickly. I told them that when they do apply for changes to the kiln production that we will need to look at all of the changes including the separator, to determine if a major modification has occurred.²⁵⁶

Mr. Hoffman's construction of this correspondence is only credible if the memo is taken completely out of context. At the time "The Ghost of Holnam" memorandum was written, January 6, 1998, many of the changes CEMEX told CDPHE it was planning to implement in the future had already been put into place. Mr. Myers did not know that. Based on the premise that these changes would be part of a future permit application, Mr. Myers told CEMEX:

I told them that I felt that it needs to be reviewed as one project, and if they were able to push it through with their high powered lawyers and separate it out, that when they come back in to deal with the second part of it, it is still going to be reviewed as one project.²⁵⁷

CEMEX never returned to permit the "second part of it" – the changes that impacted the kiln. Thus, Mr. Hofmann's contention that the "air permits issued by CDPHE to the Lyons plant allowed production increases associated with the LEP (other upstream and downstream changes)"²⁵⁸ is simply not true. The production increases associated with the kiln modifications were not put before CDPHE in the permitting process.

C. McVehil-Monnett Advised CEMEX to Permit the Kiln

Mr. Hofmann dismisses "MMA's preliminary analysis of the LEP" as being "an overly simplistic mathematical calculation with no considerations for NOx improvements."²⁵⁹ Mr. Hofmann then claims that Mr. Baugues' netting analysis and related advice to CEMEX was superseded by "more advanced advice" in a March 25, 1998 letter from Mr. Baugues to Gary Parish of LeBoeuf, Lamb, Green & MacRae.²⁶⁰ Mr. Hofmann's timing and characterization of these events are misleading, incorrect, and not cited to any specific documents.

Mr. Baugues, MMA, prepared a conventional netting analysis that showed that the Dowe Flats quarry project would trigger PSD review for NOx, based on the projected increase in raw feed to the kiln. His analysis concluded this project would increase in NOx emissions from 324

²⁵⁶ Memorandum from Dennis Myers, Re: The Ghost of Holnam, January 6, 1998, US00006499 (emphasis added).

²⁵⁷ Deposition of Dennis Myers, July 21, 2010, TR at 155:4 – 9.

²⁵⁸ Hoffman Expert Report, Summary of Opinions, p. 1-6, Opinion No. 17.

²⁵⁹ Hofmann Expert Report, Summary of Opinions, p. 1-6, Opinion No. 18.

²⁶⁰ Hofmann Expert Report, p. 5-26.

ton/yr²⁶¹ to 452 ton/yr,²⁶² depending upon baseline assumptions.²⁶³ Mr. Baugues was specifically directed by CEMEX not to disclose this analysis to CDPHE.

Q. Earlier you said that Jit Gill instructed you not to discuss the work on the kiln and the dryer with the State. Do you remember that? A. Yes. Q. What work was it that he was referring to? A. What he did not want to discuss are potential emission increases of NO_x, SO₂, VOC, and CO from the kiln. Once again, because they had no emissions offsets, that would have triggered PSD. So he did not want that discussed. Q. Did he give you justification for not discussing it with the State? A. No. I mean, it was just, "We do not want to go down that path. We're not going to discuss that with the State."²⁶⁴

Mr. Baugues faxed Mr. Lohr a handwritten summary of his draft netting analysis on March 11, 1998.²⁶⁵ This fax showed that CEMEX's then-proposed increase in clinker to 575,000 ton/yr would increase NO_x emissions by 365 ton/yr, thereby exceeding the PSD significance threshold of 40 ton/yr.²⁶⁶ Mr. Baugues concluded that CEMEX "[m]ust separate out kiln, dryer, clinker cooler in order to permit other changes."²⁶⁷

Mr. Baugues followed the fax up with a typewritten letter to Mr. Lohr on March 12, 1998 which was hand delivered in a meeting.²⁶⁸ It stated:

²⁶¹ Emissions Summary (TPY), 1996 – 97 Baseline, US00028915 at 28921 (786-98/Keith/Summary.xls) and US00000047 at 52 (**323.63 ton/yr**); Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on "Kiln and Dry Emissions," US00000065 at 80 – 82 and "Table 1" (handwritten), US00000090 (**365 ton/yr**); Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 at 38874 (**365 ton/yr**).

²⁶² Letter from Keith Baugues to Bradley Cooley, Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294; Emissions Summary (TPY), 1995 – 96 Baseline, US00028915 at 28920 (786-98/Keith/Summary.xls) and US00000047 at 51 (**451.93 ton/yr**); Deposition of Keith Baugues, July 30, 2010, TR at 158:21 – 23 ("I believe these were final calculations before we prepared the permit application.").

²⁶³ Fox Expert Report, p. 21.

²⁶⁴ Deposition of Keith Baugues, July 30, 2010, TR at 232:21 – 233:13

²⁶⁵ Deposition of Keith Baugues, July 30, 2010, TR at 91:9 – 12; Baugues Exhibit 43, Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on "Kiln and Dry Emissions," US00000065 at 80.

²⁶⁶ Fax from Keith Baugues, McVehil-Monnett Associates, Inc., to John Lohr, Southdown, March 11, 1998, attaching handwritten section on "Kiln and Dry Emissions," US00000065 at 80 – 82; "Conclusions" (handwritten), US00000062; and "Table 1" (handwritten), US00000090. *See also*, "Fax Cover Sheet," US00034559 at 34562; "Kiln and Dryer Emissions" (handwritten), US00034618 – 34620; "Table 1" (typewritten), US00034473.

²⁶⁷ Fax from Keith Baugues to John Lohr, March 11, 1998, "Conclusions" (handwritten), US00000062; Deposition of Keith Baugues, July 30, 2010, TR at 91:13 – 94:4.

²⁶⁸ Deposition of Keith Baugues, July 30, 2010, TR at 56:13 – 25; Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 (Baugues Deposition Exhibit 44).

This memo estimates emission increases for the kiln and dryer. These pose special problems because we do not have any sources to use as emission offsets for SO₂, NO_x, CO, VOC or lead.... Emission increases from all sources must remain below these significance levels or New Source Review (NSR) or Prevention of Significant Deterioration (PSD) rules come into effect.²⁶⁹

Mr. Baugues explained: “the NO_x emission increase is 368 tons [sic] per year compared with a limit of 40 tons per year. This poses the largest single impediment for repermitting this facility.”²⁷⁰ This is a significant increase in terms of PSD and would have required a PSD permit and use of best available control technology.²⁷¹ CEMEX rejected this advice because they were in a hurry to permit the quarry.²⁷² However, Mr. Baugues told CEMEX: “We need to discuss what can be done to reduce the potential emission increases for NO_x and SO₂ if we are to proceed with repermitting this facility.”²⁷³

As MMA believed the project should go through PSD review and CEMEX did not, MMA recommended that an outside law firm be retained.²⁷⁴ Mr. Baugues wrote to Mr. Parish on March 25, 1998 to seek (not give) legal advice.²⁷⁵ As explained by Mr. Baugues:

²⁶⁹ Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 – 38873 (Baugues Deposition Exhibit 44).

²⁷⁰ *Id.* at CMXLYO00038873.

²⁷¹ Deposition of Keith Baugues, July 30, 2010, TR at 97:7 – 24.

²⁷² Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, CMXLYO00038872 – 38875 (Baugues Deposition Exhibit 44); Deposition of Keith Baugues, July 30, 2010, TR at 101:2 – 102:3 (“...They didn’t have time to go through a PSD program for what they wanted to get done....”); 117:7 – 11 (“It was just not the path they wanted to chose because they wanted to get a permit quicker than the path I would have suggested for them.”).

²⁷³ Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 – 38875.

²⁷⁴ Deposition of William Monnett, November 30, 1020, TR at 23:20 – 24:20:

Q. Okay. Did there come a time when you became involved in discussions with Mr. Lohr or others at CEMEX about whether or not the plant should be included in the 1998 permitting project? A. Yes. . . . We -- again, based on my recollection, we got an outside law firm hired sometime roughly in the February 10 to 15 time frame, and we had various communications with them. We had a joint meeting, as I recall, somewhere the order of late February that involved legal as well as McVehil-Monnett and CEMEX where we laid out our initial concerns. Again, there were additional analyses that Keith did. And it all kind of culminated in a meeting that I believe happened either in late April or early May 1998. The purpose of that meeting was for McVehil-Monnett and outside counsel to make our claim as to where we thought things should go.

²⁷⁵ Letter from Keith Baugues to Gary Parish, Re: Permitting Strategy for Revisions at the Lyons Southdown Facility: MMA Project Number 786-98, March 25, 1998, CEMEX_NEIC_00008538, beginning at 8551.

A. It was, once again, the implication that I believe this project needed to go through PSD, the source did not seem to think so, and we thought maybe legal advice needed to be provided to those folks at Southdown that we were fairly certain in our conclusions and wanted them to also be sure that that was the decision they were making. Q. When you say you were certain in your conclusions, does that mean McVehil-Monnett was certain in its conclusions that the project needed PSD permitting? A. If these were part of a debottlenecking, which we thought they were, that would be the case. And at that point I still believed that was the case and that was the advice I told them, that they needed to seek a PSD permit, and we were asking them to seek legal advice to make sure they would come to the same conclusion.²⁷⁶

Contrary to Mr. Hofmann's characterization, the March 25, 1998 Baugues letter was not "more advanced advice to CEMEX" from MMA²⁷⁷ but rather an attempt by MMA to seek advice from counsel.²⁷⁸

The very next day, March 26, 1998, Mr. Baugues sent his final netting analysis²⁷⁹ to CEMEX Corporate.²⁸⁰ This analysis was contained in a series of spreadsheets named "SUMMARY.XLS – Baseline and Future Comparisons."²⁸¹ It contains netting analyses to determine whether PSD review is triggered for four cases: (1) 1995-96 baseline "all sources included"; (2) 1996-97 baseline "all sources included"; (3) 1995-96 "excludes kiln, dryer and clinker cooler"; and (4) 1996-97 "excludes kiln, dryer and clinker cooler." This spreadsheet²⁸² calculates the future potential increase in emissions due to the expansion with and without clinker production equipment (kiln, dryer, cooler) relative to two baseline periods, 1995 – 1996 and 1996 – 1997. The resulting increases in emissions are compared to PSD significance

²⁷⁶ Deposition of Keith Baugues, July 30, 2010, TR at 124:17 – 125:13 (emphasis added).

²⁷⁷ Hofmann Expert Report, Summary of Opinions, p. 1-6, Opinion No. 18.

²⁷⁸ Deposition of William Monnett, November 30, 2010, TR at 47:2 – 48:1:

Q. Can you explain to me the genesis of this document? [March 25, 1998 letter] How did Mr. Baugues come to be directed to write a letter on permitting strategy? A. I don't know specifically, but my -- I would presume that it was a request from counsel to provide some thoughts along -- thoughts regarding how we might proceed with the changes that he talks about here. . . . A. Best I can understand this, I think he's asking counsel whether or not there is a route that we can take to do that. Q Okay. Was it McVehil-Monnett's position that that could or could not be done? A. I think we were looking for legal advice as to whether it could or couldn't. We were uncertain.

²⁷⁹ Deposition of Keith Baugues, July 30, 2010, TR at 158:21 – 23 ("I believe these were final calculations before we prepared the permit application.").

²⁸⁰ Letter from Keith Baugues to Bradley Cooley, Southdown, Inc., Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294.

²⁸¹ Emissions Summary, US00000047 – 54 and US00028915 – 28923.

²⁸² Emissions Summary, US00000047 – 54 and US00028915 – 28923.

thresholds in tons/yr. These are conventional netting analyses. These spreadsheets show that CEMEX knew before it submitted its initial permit application for the quarry expansion/separator to CDPHE on April 13, 1998 that the expansion would trigger PSD review by a wide margin if all sources were included in the netting analysis. This conclusion did not change regardless of the baseline year used in this calculation. These were Mr. Baugues' final calculations, which he submitted to CEMEX after the March 25, 1998 letter to counsel.

Contrary to Mr. Hofmann's opinion, the record in this case indicates MMA did not – and has not to this day – changed their opinion that PSD review was required based on the March 25, 1998 letter to counsel nor did it change its view after MMA, Mr. Parish, and CEMEX conferred on this letter.

Q. It was McVehil-Monnett's recommendation that the plant kiln, dryer, and clinker cooler should be permitted together with the quarry -- A. True. Q. -- is that correct? A. Yes, sir. Q. Is that still McVehil-Monnett's opinion, sitting here today? A. Nothing has changed that, yes. Q. And it is still your opinion? A. Based on what I know today. Q. Okay. Based on what you know includes all those historical conversations that you've been privy to? A. Yes, sir.²⁸³

²⁸³ Deposition of William Monnett, November 30, 2010, TR at 41:20 – 42:10 (emphasis added).

APPENDIX A

References Directly Used in the Preparation of the Rebuttal Report

BOOKS

Kurt E. Peray, The Rotary Cement Kiln, Classical Publishing Co., 1986.

CASES

Puerto Rican Cement Co. v. U.S. EPA, 889 F.2d 292 (1st Cir. 1989).

DEPOSITIONS

Deposition of Andrew Jackura, September 16, 2010.

Deposition of Bradley Cooley, September 20, 2010.

Deposition of Chuck Kendrick, January 11, 2011.

Deposition of Dennis Myers, July 21, 2010.

Deposition of Gregory Wright, November 9, 2010.

Deposition of John Lohr, October 11, 2010.

Deposition of Keith Baugues, July 30, 2010.

Deposition of Kevin Kelley, December 16, 2010.

Deposition of Monica Sowders, January 5, 2011.

Deposition of Ram Seetharam, August 11, 2010.

Deposition of Ram Seetharam, August 12, 2010.

Deposition of Randal Wiley, November 11, 2010.

Deposition of Roland Hea, November 2, 2010.

Deposition of Steven Mossberg, September 8, 2010.

Deposition of William Monnett, November 30, 2010.

United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Deposition of Amarjit Singh Gill, March 18, 2008.

United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Deposition of Andrew Jackura, February 27, 2008.

DOCUMENTS

1997 Oxygen Test Spreadsheet, CMXLYO000612454.

Annual Emissions Inventory, Environmental Strategy, Lyons 2007, CMXLYO00151502.

Annual Emissions Inventory, Environmental Strategy, Lyons 2007, CMXLYO00113389.

Attachment to April 30, 1998, Application, CMXLYO00144605.

Case Study of Physical Modeling at Southdown's Lyons Plant, CMXLYO000402635.

CEMEX Lyons Facility, Estimated Clinker Production – Historical Summary, CMXLYO000628849.

CEMEX Lyons Facility, NOx Emissions – Kiln – Historical Summary, CMXLYO000628845.

CEMEX, Environmental Strategic Plan, Lyons Plant, May 2007, CMXLYO00151454.

CEMEX, Inc., Lyons Cement Plant, Capital Projects 1992 to 1999, CMXLYO00180114.

CEMEX, Lyons Plant, December 2002 MTD, CMXLYO00045217 at 45218.

Chart2, CMXLYO00038900 – CMXLYO00038902.

Colorado Department of Public Health and Environment – Air Pollution Control Division, Permit No. 98-BO-0259, September 11, 1998, CEMEX_NEIC_00000246 (New finish mill).

Colorado Department of Public Health and Environment – Air Pollution Control Division, Permit No. 98-BO-0292, September 11, 1998, CEMEX_NEIC_00000229 (New raw materials storage and handling at the plant site).

Colorado Department of Public Health and Environment – Air Pollution Control Division, Permit No. 93-BO-1414-F, September 11, 1998, CEMEX_NEIC_00000257 (Modifications to Dowe Flats Quarry).

Daily Operations Report, August 22, 1998, CMXLYO00086283.

Daily Operations Reports at CMXLYO00085234 – 95452.

Daily Production Notes for 2/26/99 – 12/17/99, CMXLYO00095456 – CMXLYO00095725.

Daily Production Notes, September 21, 1998, CMXLYO00084921.

Email from John Lohr to Dave Repasz, et. al, September 19, 2000, Re: Power Review.xls, CMXLYO00045599; CMXLYO00045547.

Email from Mike McCabe to Rick Schnarre and Philip Alsop, Re: Clinker, October 17, 2000, CMXLYO000402668.

Email from RD Smith to Naseem Khan, Re: Fuel Efficiency, November 14, 2003, CMXLYO00045243.

Emissions Summary (TPY), 1995 – 96 Baseline, US00028915 at 28920 (786-98/Keith/Summary.xls) and US00000047 at 51.

Emissions Summary (TPY), 1996 – 97 Baseline, US00028915 at 28921 (786-98/Keith/Summary.xls) and US00000047 at 52.

Emissions Summary, US00000047 – 54; US00028915 – 28923.

Fax from Keith Baugues to John Lohr, March 11, 1998, “Conclusions” (handwritten), US00000062.

Fax from Keith Baugues to John Lohr, March 11, 1998, “Fax Cover Sheet,” US00034559 at 34562.

Fax from Keith Baugues to John Lohr, March 11, 1998, “Kiln and Dryer Emissions” (handwritten), US00000065 at 80 – 82; US00034618 – 34620.

Fax from Keith Baugues to John Lohr, March 11, 1998, “Table 1” (handwritten) US00000090.

Fax from Keith Baugues to John Lohr, March 11, 1998, “Table 1” (typewritten) US00034473.

Handwritten Notes from the State, 1998, US00006339 – 6345; CMXLYO00015627 at 15629 – 15638.

Herman H. Tseng & Philip A. Alsop, Apparatus for Oxygen Enrichment of Cement Kiln System, U.S. Patent No. US 6,688,883 B2, February 10, 2004, Patent Application, February 10, 2004, CMXLYO000628933.

Herman Tseng & John W. Lohr, Oxygen Enrichment to Calciner Firing, CMXLYO000612639.

Herman Tseng & John W. Lohr, Oxygen Enrichment, International Cement Review, May 2001, CMXLYO000612451.

John Lohr, Plant Production Bottlenecks, Lyons, Colorado, January 14, 2004, CMXLYO000044998.

Letter from David Repasz, Southdown, Inc., to Kenneth M. Hitcho, Humboldt Wedag, Inc., June 12, 1998, Re: Letter of Intent for Supply to Finish Grinding Separator System for Lyons, Colorado Plant Authorized Spending, CEMEX_NEIC_00003381.

Letter from John Lohr, Plant Manager, CEMEX, to Ram Seetharam, CDPHE –APCD, April 29, 1998, CEMEX_NEIC_00008478.

Letter from Keith Baugues to Bradley Cooley, Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294.

Letter from Keith Baugues to Bradley Cooley, Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294.

Letter from Keith Baugues to Gary Parish, Re: Permitting Strategy for Revisions at the Lyons Southdown Facility: MMA Project Number 786-98, March 25, 1998, CEMEX_NEIC_00008538, beginning at 8551.

Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872.

Lyons Capacity Increase, October 17, 1997, CMXLYO00182398.

Lyons Clinker Production History, May 17, 2000, CMXLYO00045592.

Lyons Kiln History and Forecast (1992 – 1997), CMXLYO00155750.

Lyons Kiln Slowdowns in 1999 (June-December), CMXLYO00157286.

Lyons Plant – Monthly Report of Operations, August 1999, CMXLYO00042970.

Lyons Plant – Monthly Report of Operations, December 1995, CMXLYO00047047.

Lyons Plant – Monthly Report of Operations, December 1999, CMXLYO00043081.

Lyons Plant – Monthly Report of Operations, January 1999, CMXLYO00043386.

Lyons Plant – Monthly Report of Operations, May 2001, CMXLYO00043894.

Lyons Plant Cost Reduction Meeting, March 18 & 19, 1999, CMXLYO00045695.

Lyons Plant Expansion – Phase I, Project Scope – Revised 12/15/97,
CEMEX_NEIC_00003299, beginning at 3301.

Lyons Plant Expansion, Project Scope – Revised 9/21/98, CMXLYO00155865.

Lyons Plant Pyroprocess Training, CMXLYO00186916.

Lyons Plant, 2000 Plan, January 14, 2000, CMXLYO00415808.

Lyons Plant, August 28, 1998, CMXLYO00155783.

Memorandum from Dennis Myers, Re: The Ghost of Holnam, January 6, 1998,
US00006499.

Memorandum from Gregg St. Clair to Dave Repasz, March 26, 1998, Re: Plant
Expansion Project Update, CMXLYO00155748.

Memorandum from Gregg St. Clair to Dave Repasz, November 19, 1997, Re: Lyons
Plant Expansion – Phase I, CEMEX_NEIC_00003291.

Memorandum from Gregg St. Clair to Dave Repasz, Re: Plant Expansion Project Update,
CEMEX_NEIC_00003247.

Memorandum from John Lohr to Alan Redeker, July 21, 1998, Re: Potential Permit
Impacts on Plant Expansion Project, CEMEX_NEIC_00003421.

Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, January 4, 1998, Re:
Lyons Kiln Production and Oxygen Use, CMXLYO00039059; US00000020.

Memorandum from Karen Krenek to Norris Widener, November 18, 1998, Re: Approved
AFE, US00000454; CEMEX_NEIC_00003449.

Monthly Financial Report, April 30, 1997, CMXLYO00049893.

Monthly Financial Report, January 31, 1998, CMXLYO00050091.

Monthly Financial Report, March 31, 1997, CMXLYO00049696.

Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155868.

PowerPoint, CEMEX, Inc., Lyons Cement Plant, CMXLYO00150942;
CMXLYO00151197.

PowerPoint, Southdown, Inc., Meeting with Colorado Department of Health and Environment, APCD, March 6, 1998, US00000594.

Project Outline for Oxygen Enrichment Test at Lyons – March 1997, CMXLYO00402688; CMLXYO00612385.

Riser Coal Firing, Lyons, Colorado, April 9, 2002, CMXLYO00045284.

Southdown, Inc. Lyons Plant, Process Additions, Steel Slag and Oxygen Injection, March 7, 2000, CMXLYO00045630; CMXLYO00045569.

Southdown, Inc., Authorization for Expenditure, November 18, 1998, US00000027.

Southdown, Inc., Construction Permit and Permit Modification Application, April 30, 1998, CMXLYO00015334, beginning at 15582 (Overview of all permits).

Southdown, Inc., Lyons Cement Plant, Air Pollution Emission Notice, December 6, 1995, US00006692.

Southdown, Inc., Lyons Cement Plant, Clinker Production, US00000583.

Technical Services Group, Activity Report, September 1, 1996, US00078597, beginning at 78766.

Text and Figures for FCT AFE, CMXLYO00045595.

Untitled Spreadsheet (2000 – 2007), CMXLYO00117455.

Variance Analysis, August 2003, CMXLYO00045237.

Variance Analysis, July 2003, CMXLYO00045236.

Variance Analysis, November 2003, CMXLYO00045245.

PAPERS

Gerald L. Young, NO_x Formation in Rotary Kilns Producing Cement Clinker Applicable NO_x Control Techniques and Cost Effectiveness of these Control Techniques, IEEE Environmental, Safety and Maintenance Working Group, Jacksonville Conference May 2002.

Letter from Cheryl L. Newton, Director, Air and Radiation Division, to Keith Baugues, Assistant Commissioner, Officer of Air Quality, Indiana Department of Environmental Management, April 4, 2011, p. 2. Available at:
<http://www.epa.gov/region7/air/nsr/nsrmemos/atpanet.pdf>.

Memorandum from Al Linero to Howard Rhodes, Re: Southdown, Inc., 0530010-003-AC and PSD-FL-233, Modification of Existing Construction Permit, Final Permit, June 17, 1997, Attaching Notice of Final Permit, Final Determination, and Air Construction Permit, pdf 32 (BACT limits) and Table 1-2.

Memorandum from John B. Rasnic, Director, Stationary Source Compliance Division, Office of Air Quality Planning and Standards, to George T. Czerniak Chief, Air Enforcement Branch, Region V, Re: Applicability of New Source Review Circumvention Guidance to 3M – Maple Minnesota, June 17, 1993. Available at: <http://www.epa.gov/region7/air/nsr/nsrmemos/maplwood.pdf>.

New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting, Draft, October 1990.

U.S. Environmental Protection Agency, Alternative Control Techniques Document – NOx Emissions from Cement Manufacturing, Report EPA-453/R-94-004, March 1994. Available at: <http://www.epa.gov/ttn/catc/dir1/cement.pdf>.

U.S. Environmental Protection Agency, AP-42, Fifth Edition, January 1995, Compilation of Air Pollutant Emission Factors, Vol. 1, Stationary Point and Area Sources.

United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Gerald L. Young, Penta Engineering, NOx Formation in Rotary Kilns Producing Cement Clinker Applicable NOx Control Techniques and Cost Effectiveness of these Control Techniques, IEEE Cement Industry Technical Conference, 2002, USPROD01-003473.

United States v. CEMEX California Cement, LLC, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW: KHD Humboldt Wedag, Operating Instructions for Rotary Kiln with Single-String Cyclone Preheater PR 73(76)52 PYROCLON with PYROTOP Coal Dust Firing, USPRODCONT-0000221.

Zephyr Environmental Corporation, Summary of Control Techniques for Nitrogen Oxides, Portland Cement Association, PCA R&D SN3049, 2008, p. 13. Available at: <http://www.cement.org/bookstore/download.asp?mediatypeid=1&id=16594&itemid=SN3049>.

REGULATIONS

40 CFR 52.21(b)(4) (1995) (presently recodified as 40 CFR 52.21(b)(23)(i)).

56 Fed. Reg. 27630, June 14, 1991.

Colorado Air Quality Control Commission, Common Provisions, I.G (1986).

Colorado Air Quality Control Commission, Part A, I.B.44 (1997).

REPORTS

Expert Report of Gerald L. Young, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011.

Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011.

Expert Report of Kevin Davis, Ph.D, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011.

Expert Report of Phyllis Fox, Ph.D, PE, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011.

Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011.

Rebuttal Expert Report of Bradley Venner, Ph.D., United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011.

Rebuttal Expert Report of Peter R. Westlin, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 15, 2011.

Rebuttal Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, June 24, 2011.